

Senior Thesis

Abstract

Minnesota felony sentencing data can teach students how to reason responsibly with criminal-justice statistics, but only if every pattern is interpreted through the criminal-justice funnel, the dataset's administrative origins, the unit of analysis, and the difference between description and explanation.

Introduction: Reading Sentencing Data Without Losing the Plot

Opening Problem: Crime Data Feels Obvious, But Isn't

Purpose of this section

Begin with the central public problem: people regularly encounter claims about crime, punishment, race, gender, severity, leniency, and “what courts are doing,” but criminal-justice data is easy to misread because every number comes from a system of selection.

Core move

Open with a concrete tension:

- Crime statistics are politically and emotionally powerful.
- Sentencing statistics appear precise.
- But precision is not the same thing as meaning.
- A number can be accurate and still be misinterpreted.

Possible thesis-facing language

The introduction should establish that this document is about learning to read sentencing data carefully, not merely learning to produce tables.

Key claim

The Minnesota felony sentencing dataset is valuable because it is detailed, large, and systematic, but it is also limited because it records only cases that reached felony sentencing. The draft already makes this the foundation of Module 1: the dataset contains roughly 294,000 felony

sentencing records from 2001 through 2019, and those records sit at the bottom of the criminal-justice funnel.

What This Project Is

Purpose of this section

Define the document's identity.

It is three things at once

1. **A data-literacy curriculum**

Students learn how to filter, summarize, compare, and communicate findings.

2. **A sentencing-system primer**

Students learn how Minnesota's felony sentencing system produces the variables they analyze.

3. **A methodological caution manual**

Students learn how not to confuse description with explanation.

Clarify the genre

This is not a conventional empirical thesis that asks one research question and answers it once. It is a guided curriculum for teaching students how to ask answerable descriptive questions using felony sentencing data.

Possible language

This project treats the Explorer not merely as a tool for displaying data, but as a classroom for learning how criminal-justice statistics acquire meaning.

The Dataset and the Explorer

Purpose of this section

Briefly introduce the Minnesota Sentencing Explorer and the dataset before the modules begin.

Include

- The dataset covers Minnesota felony sentencing records.
- The relevant years are 2001–2019.
- The approximate size is 294,000 records.
- Each row is a sentencing record, not one person and not one crime.
- The Explorer allows students to filter, summarize, revert, clear, and cross-tabulate the data.
- The dataset is administrative data, produced by the sentencing system rather than designed as a research survey.

Important distinction

The Explorer makes analysis accessible, but accessibility creates risk: students can generate comparisons before they understand what those comparisons mean.

Possible language

The interface lowers the technical barrier to analysis, but the intellectual barrier remains. The purpose of the course is to build that intellectual discipline.

The Central Problem: Sentencing Data Is Not Crime Data

Purpose of this section

Preview the document's foundational interpretive rule.

Core ideas to introduce

- The criminal-justice funnel.
- The dark figure of crime.
- Administrative data.
- Sellin's dictum.
- Sentencing records as system outputs.
- Description versus explanation.

Key claim

This dataset is excellent for studying how Minnesota recorded and structured felony sentencing decisions. It is not a direct measure of how much crime occurred in Minnesota.

Possible language

A felony sentence is not the beginning of a criminal event; it is the end of a long institutional process. By the time a case becomes a row in the dataset, countless other events have already disappeared from view.

The Pedagogical Problem: Students Need More Than Statistics

Purpose of this section

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Explain why the course teaches legal and criminological context alongside data skills.

Core claim

Students cannot responsibly interpret sentencing data by learning statistics alone.

Explain why

To read the dataset, students need to understand:

- what a felony is;
- how offense classifications work;
- why Minnesota uses sentencing guidelines;
- what criminal-history scores do;
- what presumptive sentences are;
- how stayed and executed sentences differ;
- why a raw group difference may reflect offense mix, criminal history, selection, charging, plea bargaining, judicial discretion, or unrecorded facts.

Possible language

The same table can support a careful descriptive statement or a reckless causal claim. The difference is not the table. The difference is the reader.

The Main Thesis

Purpose of this section

State the project's controlling argument clearly.

Working thesis

This document argues that Minnesota felony sentencing data can teach students to reason responsibly about criminal-justice patterns, but only if every finding is interpreted through four recurring questions:

1. **Where in the criminal-justice funnel does this number sit?**
2. **What is the unit of analysis?**
3. **What has been filtered, excluded, or collapsed before the number appears?**
4. **Is the claim descriptive, or is it trying to explain cause?**

Stronger version

The project's central claim is that data literacy in criminal justice is not simply the ability to compute statistics. It is the ability to know what a statistic can bear.

Roadmap

Purpose of this section

Preview the five-unit structure so the reader understands the progression.

Unit I: Foundations

Students learn what the data is, how the Explorer works, and how to read descriptive statistics.

This unit builds the basic grammar of the course: funnel, unit of analysis, filtering, mean, median, standard deviation, percentages, and N.

Unit II: The Machine

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Students learn how the sentencing system creates the categories and numbers they analyze. This includes offense classification, the Minnesota Offense Code, the sentencing grid, criminal-history score, and the meaning of time.

Unit III: Reading Patterns Responsibly

Students learn how patterns change over time, how to hold variables constant, why small numbers mislead, and how to interpret sex differences without jumping from disparity to cause.

Unit IV: The Big Question

Students apply the full method to race and sentencing disparity. The emphasis is on distinguishing disparity from discrimination and identifying what the dataset can and cannot establish.

Unit V: Capstone

Students design and communicate their own descriptive investigation. The filter history becomes the methods section, and the final report must defend its choices, limitations, and language.

What the Document Will Not Do

Purpose of this section

Establish intellectual honesty early.

Important exclusions

This project does not:

- measure all crime in Minnesota;
- estimate the true amount of offending;

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- determine whether any specific sentence was fair;
- predict what sentence a future defendant will receive;
- prove or disprove discrimination by itself;
- replace more advanced causal research designs;
- treat the dataset as a complete account of defendants' lives, case facts, charging negotiations, or judicial reasoning.

Possible language

The refusal to overclaim is not a weakness of the project. It is the project's central discipline.

Outcomes

Purpose of this section

End the introduction by naming what students should be able to do.

By the end, students should be able to

- produce a descriptive finding;
- explain how they got it;
- identify the relevant filters;
- check whether N is large enough to trust;
- choose mean or median intentionally;
- compare groups using percentages when appropriate;
- distinguish raw gaps from conditional gaps;

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- describe what remains unknown;
- write a finding without causal leakage.

Possible closing sentence

The goal is not to make students suspicious of every number. It is to make them worthy of using numbers that matter.

Unit I: Foundations

Module 1: What This Data Is (and Is Not)

Situating the Minnesota Felony Sentencing Dataset

Every analytic task that follows, whether filtering, summarizing, or comparing groups, depends on a prior question: what is the dataset students are exploring, and what can it honestly tell them? That question is not a formality. Numbers about crime are among the most frequently misread statistics in public life. In 23 of 27 national surveys conducted since 1993, a majority of Americans reported believing that crime was rising, even across long periods in which recorded crime was falling (Gramlich, 2024). A statistic is only as trustworthy as the reader's understanding of where it came from and what it leaves out (Best, 2001).

This chapter therefore begins where careful analysis must begin: with provenance. It traces the Minnesota felony sentencing dataset back to its institutional origin, asking who created it, why it was created, and where in the life of a criminal case the record was produced. The purpose is not

merely to introduce the data. It is to make clear which questions the roughly 294,000 felony sentencing records covering 2001 through 2019 can answer, and which questions they cannot.

Administrative Data: Built to Run a System, Not to Answer a Question

The dataset is administrative data: information generated as a by-product of running an organization rather than collected to answer a research question (Connelly et al., 2016). Courts and the Minnesota Sentencing Guidelines Commission record sentences because the state must document and monitor how people are sentenced. They do not record them because a researcher first designed a study.

That origin is both a strength and a limitation. Administrative records can cover nearly every case, span decades, and avoid the sampling error that accompanies surveys. Yet because the data was shaped by the needs of the system that produced it, using it for research requires reconstructing how each field was defined and recorded. The categories, rules, and exceptions were chosen by administrators, not by the analyst (Connelly et al., 2016).

The first habit, then, is to treat every column as an administrative decision before treating it as a measurement. The question is not only what value appears in the field. The question is why the system would record that value in that way. That discipline prevents many of the mistakes that follow when institutional records are read as if they were neutral windows onto social reality.

The Source: The Minnesota Sentencing Guidelines Commission

In 1978, the Minnesota Legislature created the Minnesota Sentencing Guidelines Commission, and in 1980 Minnesota became the first state to sentence felonies under presumptive guidelines developed by such a commission (Frase, 2005; Minnesota Sentencing Guidelines Commission, n.d.). The guidelines derive a presumptive sentence for each case from two facts: the severity of

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the offense and the sentenced person's criminal-history score. The structure was intended to make punishment more uniform and principled (Frase, 2013).

To monitor whether courts follow the guidelines, the Commission maintains a record of every person convicted of a felony in adult court and sentenced under them. Those records are assembled from probation officers' sentencing worksheets and court data (Minnesota Sentencing Guidelines Commission, 2024). The present dataset is a slice of that monitoring system. For scale, annual felony case volumes have ranged from roughly 5,500 in 1981 to a peak of 18,288 in 2017 (Minnesota Sentencing Guidelines Commission, 2023).

This provenance matters for interpretation. The data exists to answer the Commission's question: are sentences following the guidelines? It was not designed to measure how much crime Minnesota experiences, or why. The dataset is powerful when it is used for the purpose its structure can support. It becomes fragile when it is asked to answer questions it was never built to answer.

The Criminal-Justice Funnel

A sentence is the last event in a long sequence. For a felony sentence to exist, a crime must have been committed, noticed, reported, recorded by police, and investigated. An arrest must have been made, charges must have been filed, guilt must have been established, and a felony-level sentence must have been imposed.

Criminologists often describe this progression as a funnel: many events enter at the top, and only a fraction reach the bottom. The conception of criminal justice as a connected system, with cases dropping out at each stage, was articulated by the President's Commission on Law Enforcement and Administration of Justice (1967).

The Minnesota felony sentencing dataset sits at the very bottom of that funnel. It contains only cases that survived every prior stage and ended in a felony sentence. It therefore says nothing directly about crimes no one reported, reported crimes that produced no arrest, arrests that produced no charge or conviction, or offenses resolved as misdemeanors or dismissed. Each of those is a stage the dataset never observes. The record begins late in the story.

The Dark Figure of Crime

Criminologists call the crime that never appears in official records the dark figure of crime: conduct treated as crime but registered in no agency's statistics (Biderman & Reiss, 1967). The dark figure is not small. In the National Crime Victimization Survey, only about 42% of violent victimizations were reported to police in 2022, meaning nearly six in ten went unreported (Morgan & Thompson, 2023).

Nor is unreported crime a random sample of all crime. It tends to skew toward less serious incidents, smaller losses, and less injury (Skogan, 1977). The crimes that move through the funnel toward a felony sentence are therefore systematically different from those that do not.

Before a single row of the Minnesota sentencing dataset was recorded, most crime had already dropped out of view. It had also dropped out selectively. No filter applied later can recover what the system never recorded.

Sellin's Dictum: Distance From the Act

In 1931, criminologist Thorsten Sellin articulated a principle now known as Sellin's dictum: the farther a statistic is collected from the criminal act itself, measured in procedural steps, the less reliable it is as an index of crime (Sellin, 1931). Police reports sit closer to the event than arrests. Arrests sit closer than convictions. Convictions sit closer than sentences. Sentences lie as far down the procedural chain as it is possible to go.

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By Sellin's rule, felony sentencing data is an excellent record of one thing: what the system decided to do. It is a weak proxy for another: how much crime actually occurred. This distinction reframes the entire enterprise. The roughly 294,000 records provide a rich and reliable account of Minnesota sentencing decisions. Read as a map of crime itself, they are among the weakest measures available.

The mistake is understandable. Sentencing records are official, detailed, and numerically precise. But precision is not the same as proximity. Almost every error in interpreting the data begins by confusing a downstream sentencing record with an upstream criminal event.

How the Nation Measures Crime, and How Minnesota's Data Differs

Because no single source captures all crime, the United States relies on two national systems that sit at different points in the criminal-justice funnel. The FBI's Uniform Crime Reporting program, now transitioning to the more detailed National Incident-Based Reporting System, counts crimes recorded by police (Federal Bureau of Investigation, n.d.). The Bureau of Justice Statistics' National Crime Victimization Survey instead asks a national sample of households about crimes they experienced, including crimes never reported to police (Bureau of Justice Statistics, n.d.-b).

The two systems disagree by design because they measure different slices of the funnel. The Minnesota data is neither of these. It is a court and sentencing record that sits below both in the funnel.

The proper question before comparing a finding from this dataset to a national crime statistic is therefore not whether the numbers look similar. It is whether they measure the same stage of the process. Usually, they do not.

The Unit of Analysis: What One Row Represents

A dataset is only as clear as its unit of analysis. Before a total can be trusted, the reader must know what one row represents.

Here, one row is one sentencing record. The Commission's counting rules define that record precisely: when a person is sentenced in the same county for several offenses within a 30-day window, those offenses are combined into a single record described by the most serious offense (Minnesota Sentencing Guidelines Commission, 2018).

Two consequences follow, and both are easy to miss. First, one row is not one person. An individual sentenced in 2004 and again in 2015 occupies two rows because the data counts sentencing events rather than individuals. This is the same distinction federal court statistics draw between a case and a defendant (Bureau of Justice Statistics, n.d.-a).

Second, one row is not one crime. Secondary offenses within a record are collapsed into the most serious offense, and, as the funnel shows, most crimes never produce a sentencing record at all.

Every total in the dataset should therefore be read not as a count of people or crimes, but as a count of sentencing records. The full dataset comprises roughly 294,000 such records described across more than 100 administrative fields. Each field is a category the Commission chose to record. Each is also a reminder of the many things the dataset does not know.

What the Dataset Excludes

Combining the funnel with the counting rules makes the dataset's absences visible. It excludes unreported crime, the dark figure that constitutes the majority of offenses (Biderman & Reiss, 1967). It excludes crimes that produced no felony conviction, including cases dropped, dismissed,

acquitted, or resolved as misdemeanors. It excludes secondary offenses, which are folded into a record's most serious charge (Minnesota Sentencing Guidelines Commission, 2018).

It also excludes most white-collar and regulatory harm, which rarely yields a felony sentence, an offense that by definition may carry a term of at least a year. It excludes context about people's lives, because the monitoring system does not maintain data on sentenced persons' social or economic circumstances (Minnesota Sentencing Guidelines Commission, 2024).

A final caution belongs here. This is an educational dataset, not a prediction engine. It describes sentences that were imposed. It is not a calculator for the sentence a future case should or will receive. The gap between the guideline-recommended sentence and the sentence actually pronounced is itself a subtle matter, taken up in a later chapter.

Why a Pattern Is Not a Verdict: Description Versus Explanation

A sentencing record is the end product of a long chain of human decisions: victims who report or do not, police who arrest or do not, prosecutors who charge or bargain, and judges who sentence. Every stage filters which cases arrive at the bottom (Klepper et al., 1983). Studying only sentenced cases therefore means studying a selected, non-random sample of everything that happened (Bushway et al., 2007).

When two groups exhibit different average sentences, the finding is a description of the data. It is not, by itself, an explanation of the cause. Correlation does not establish causation. A landmark review of sentencing research concluded that even after accounting for the two strongest legal factors, offense severity and criminal history, most of the variation in sentences remains unexplained. It also concluded that observed group differences cannot by themselves be read as proof of discrimination (National Research Council, 1983).

Descriptive differences are real and worth identifying. Explaining why one exists is a separate and harder task. The proper reading of a cross-tabulation showing that offenses in one category receive longer average presumptive sentences than those in another is therefore disciplined and limited: a descriptive difference exists in the data, and the reasons behind it require further analysis.

The Stakes: Why Careful Reading Matters

The insistence on caution has a public rationale. Crime statistics are unusually easy to misuse, and the consequences play out in policy and public opinion. Despite a long-term decline in recorded violent crime since the early 1990s, with the rate falling by roughly half, large majorities of Americans have repeatedly told pollsters that crime is worsening (Gramlich, 2024).

People reason confidently from numbers they have not examined. That is precisely the failure data literacy is meant to prevent (Best, 2001). Three questions guard against it: Where in the funnel does this number sit? What is its unit of analysis? Is a pattern being described or explained?

Summary

The core claims of this chapter can be stated compactly. The Minnesota felony sentencing dataset is administrative data: a by-product of running the state's sentencing system rather than a study designed to measure crime. This is the distinction between found and made data. Both involve deliberate thought, but for made data the research question shapes the collection, while a monitoring dataset like this one is structured around the needs of the system that produced it (Connelly et al., 2016).

The data sits at the bottom of the criminal-justice funnel, containing only cases that survived every earlier stage and ended in a felony sentence (President's Commission on Law Enforcement and

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Administration of Justice, 1967). Most crime lies in the dark figure and never enters the funnel at all. It is also absent non-randomly (Biderman & Reiss, 1967; Skogan, 1977).

By Sellin's dictum, the dataset is an excellent record of decisions and a weak proxy for crime (Sellin, 1931). One row is one sentencing record, not one person and not one crime (Minnesota Sentencing Guidelines Commission, 2018). A pattern in the data is a description, not an explanation. Causes require separate and careful analysis (National Research Council, 1983).

The core lesson of this chapter is that the dataset records felony sentencing decisions, not crime itself. The mistake it prevents is allowing a downstream administrative record to mean more than the system that produced it can support. From this point forward, every use of the dataset should carry three questions: where in the funnel the number sits, what unit of analysis it describes, and whether the claim being made is descriptive or explanatory.

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Module 2: Driving the Explorer

Filtering, Summarizing, and Cross-Tabulating the Sentencing Data

A dataset does not become meaningful simply by virtue of being large. It becomes meaningful when the reader learns how to ask disciplined questions of it. The Minnesota Sentencing Explorer is the tool through which those questions are posed in this course. It does not explain sentencing by itself. It organizes felony sentencing records so that students can learn what a claim describes, what population it applies to, and what limits attach to it.

This module introduces the five operations that make the Explorer usable: filtering, reading column summaries, reverting, clearing, and cross-tabulating. These operations are technical in form, but methodological in consequence. Each one changes, preserves, or compares the population being described. The first habit is therefore not to look for a striking number. The first habit is to ask what rows are currently in view.

The dataset used in this thesis contains 294,467 Minnesota felony sentencing records from 2001 through 2019 (Minnesota Sentencing Explorer, n.d.). That number is close to the public annual case totals reported by the Minnesota Sentencing Guidelines Commission (MSGC), but it should not be treated as interchangeable with those public totals without checking the unit of analysis. Public reports generally summarize felony cases. The Explorer file used here is organized as sentencing records. As the previous module established, one row is one sentenced count. It is not necessarily one person, and it is not one crime committed.

That distinction matters because every statistic the Explorer reports is calculated over rows. It does not describe all criminal conduct in Minnesota. It does not describe all arrests, charges,

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dismissals, or plea negotiations. It describes records that reached felony sentencing. Minnesota law gives the MSGC a formal role in collecting, analyzing, and disseminating information about sentencing practices, imprisonment, alternatives to imprisonment, plea bargaining, and related criminal-justice matters (Minn. Stat. § 244.09, 2025). The dataset therefore sits at the bottom of a long institutional chain. By the time a case becomes a row, police decisions, prosecutorial discretion, plea bargaining, conviction, and sentencing have already shaped what the record can show.

Three variables recur throughout this module. The variable *sentyear* records the year in which the sentence was imposed. The variable *time* records presumptive sentence duration in months. That language is deliberate: MSGC defines presumptive duration as the recommended fixed sentence length in months found in the appropriate grid cell (MSGC, n.d.-a). The variable *moc1* records the first character of the Minnesota Offense Code as it appears in the Explorer file. In this dataset, *moc1* = A functions as the assault branch, and *moc1* = D functions as the drug-offense branch. These variables are simple enough to operate, but they are not simple enough to ignore. Each is an institutional record carrying the meaning assigned to it by the sentencing system.

The Starting Point: The Unfiltered Dataset

When the Explorer is opened with no filters applied, every row in the dataset is in view. The most important number on the screen is N, the count of rows currently included in the view. Before a statistic can be trusted, the reader must know how many rows it describes.

In the unfiltered Explorer dataset, N = 294,467 (Minnesota Sentencing Explorer, n.d.). That number is not a count of all crimes in Minnesota from 2001 through 2019. It is not a count of all people

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sentenced. It is the number of felony sentencing records included in the file. The arithmetic is broad. The population is specific.

This first view is useful precisely because it is so easy to misread. A full-dataset summary appears to describe “Minnesota sentencing.” The honest claim is narrower: it describes the sentencing records included in this dataset. The difference between those two statements is the difference between using data and being used by it.

Filtering

A filter keeps only the rows that match a condition and hides the rest. The underlying records are not deleted. The view has changed, not the dataset.

The Explorer allows the standard comparison operations: equals, does not equal, greater than, greater than or equal to, less than, and less than or equal to. In the Explorer’s notation, those comparisons appear as eq, ne, gt, ge, lt, and le. A filter such as *sentyear* >= 2015 keeps only sentencing records from 2015 through 2019. That filter narrows the dataset from 294,467 records to 87,597 records (Minnesota Sentencing Explorer, n.d.). The number has changed because the population has changed.

The second principle is that filters chain. A new filter does not return to the full dataset and begin again. It applies only to the rows that survived the previous filter. The Explorer records that chain as a filter history. That history matters because it is the record of how the current population was produced.

A filtered view is therefore not merely a smaller dataset. It is a claim with boundaries. If the Explorer is filtered to *sentyear* >= 2015, any statistic that follows describes sentencing records from 2015

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through 2019, not all records in the dataset. A filter changes the population. Every interpretation must follow that change.

This is the first point where software becomes method. The filter looks like a button. It behaves like a question: how does the pattern look when only this population remains?

Reading a Column Summary

Once a population is in view, the Explorer can summarize a column within that population. For a numeric column such as time, the summary includes descriptive statistics, especially the mean and the median.

The mean is the arithmetic average: all values are summed and divided by N. The median is the middle value after the values are ordered, with half of the rows below it and half above it. Both numbers describe the current filtered view. They do not answer the same question.

The distinction matters because sentencing data often contains a long right tail. Most felony sentencing records may cluster around shorter presumptive durations, while a smaller number of records carry very long presumptive durations. NIST explains that measures of spread and center are affected by tail behavior, and that histograms are useful precisely because they reveal both the center of a distribution and the behavior of its tails (National Institute of Standards and Technology [NIST], n.d.-a). The mean moves with the tail. The median describes the middle of the rows.

In the unfiltered Explorer dataset, the mean of time is 27.20 months, while the median is 17 months among records with nonmissing time values (Minnesota Sentencing Explorer, n.d.). That gap is not a software error. It is a clue about distributional shape. The mean is higher because longer presumptive durations pull it upward. The median remains closer to the typical record.

At this stage, the task is not to decide that one statistic is always better. The task is to keep them distinct. A claim about the “average” presumptive sentence is incomplete until the reader knows whether the claim refers to the mean or the median.

Chaining Filters and the Offense-Code Drill-Down

The same rule applies when the Explorer is used to examine offense categories. Because filters stack, adding a second filter narrows the population that survived the first. If the first filter is *sentyear* >= 2015, and the second filter is *moc1* == A, the resulting view contains only records sentenced from 2015 through 2019 whose *moc1* value places them in the assault branch of the Explorer code structure. In the uploaded file, that two-filter view contains 12,951 records (Minnesota Sentencing Explorer, 2026).

The Minnesota sentencing system itself is structured around offense classification and criminal-history score. MSGC explains that the sentencing grids use offense severity on the vertical axis and criminal-history score on the horizontal axis; the intersection indicates the recommended sentence length and whether the sentence is presumptively prison or stayed (MSGC, n.d.-b). The Explorer’s offense-code drill-down is therefore not just a convenient sorting device. It is a way of seeing one layer of the classification system through which sentencing records are organized.

The Minnesota Offense Code receives special treatment in the Explorer because it can be examined one character at a time. The variable *moc1* identifies a broad branch of offense coding. More specific offense categories can be reached through additional characters of the code. A drill-down should be read as a filter, not as a separate kind of operation. It joins the same filter history and narrows the same population.

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If `moc1 == A` is added after `sentyear >= 2015`, the Explorer now describes assault-branch sentencing records among records sentenced from 2015 through 2019. The order of filters may not always change the final set of rows, but the history remains pedagogically useful because it shows how the student arrived at the view.

Here is a task a student could undertake with the Explorer to understand chained filtering. The student should clear the Explorer, apply `sentyear >= 2015`, record N, then add `moc1 == A` and record the new N. This task was chosen because it makes the population change visible in two steps: first by year, then by offense classification. What the student should notice is that the second filter applies only within the population left by the first. What the task is meant to teach is that every filter narrows the claim.

Revert and Clear

Because the Explorer stores filters as a history, undoing a filter can be precise. Revert removes the most recent filter and returns the view to the previous state. Clear removes the entire filter history and returns the Explorer to the unfiltered dataset.

Neither operation changes the underlying data. Revert and clear change the view. That distinction is small in software terms but large in methodological terms. The Explorer is not rewriting the sentencing records. It is changing which records are being summarized.

A useful check follows from this structure. If a student filters to `sentyear >= 2015`, records N, then adds `moc1 == A`, and then reverts, the Explorer should return to the earlier `sentyear >= 2015` view. The N should match the value recorded before the assault filter was added. The equality is not just a software convenience. It confirms that the population has been reconstructed.

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This is the chain metaphor made visible. Each filter is a link. Revert removes the most recent link.

Clear breaks the chain entirely and returns the reader to the full dataset.

Cross-Tabulation

A cross-tabulation compares a numeric variable across two grouping variables at the same time. It splits the current view into cells and reports descriptive statistics within each cell. In this course, a common cross-tabulation might place sentyear across the columns, moc1 down the rows, and summarize time inside each cell.

The tempting reading is that a cross-tabulation is a new analytical object. The better reading is simpler: each cell is a filtered view the student could have built by hand. The cell where assault intersects with 2018 describes records with `sentyear == 2018` and `moc1 == A`. In the uploaded Explorer file, that view contains 2,628 records; among records with nonmissing time, the mean time is 21.03 months and the median is 18 months (Minnesota Sentencing Explorer, 2026).

This is the idea that demystifies the cross-tabulation. It is not a magic table. It is a compact array of filtered populations. Each number still has an N, a denominator, and a set of limits.

Here is a task a student could undertake with the Explorer to understand cross-tabulation. The student should create a cross-tabulation with sentyear as one grouping variable, moc1 as the other grouping variable, and time as the numeric variable. The student should then choose one cell, such as assault in 2018, and manually reproduce that cell by clearing the Explorer and applying `sentyear == 2018` followed by `moc1 == A`. This task was chosen because it turns the cross-tabulation from a table into a set of visible filter chains. What the student should notice is that the statistic in the cell matches the statistic in the corresponding manually filtered view. What the task is meant to teach is that a cross-tabulation is an efficient way to display many filtered views at once.

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One caution belongs here from the outset. A difference between two cells is a description of the data. It is not an explanation of why the difference exists. If the mean presumptive duration for assault records differs between two years, the Explorer has shown a descriptive gap across sentencing records. It has not identified the cause of that gap. The pattern may be real. The explanation remains a separate question.

Summary

The core lesson of this module is that the Explorer's basic operations are also habits of disciplined interpretation. A filter narrows the current view and joins the history. A column summary reports descriptive statistics for the rows currently in view. Revert steps back one filter. Clear returns to the unfiltered dataset. A cross-tabulation displays many filtered views at once.

The mistake this module prevents is treating Explorer output as free-floating fact. The Explorer computes numbers for a population. The reader's task is to name that population before interpreting the number. From this point forward, every statistic should prompt the same first question: what rows does this describe? The next module extends that discipline to the descriptive statistics themselves.

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Module 3: What the Numbers Say

Reading Descriptive Statistics

A statistic can look more authoritative than it is. A mean carried to two decimal places, a percentage displayed neatly in a table, or a standard deviation printed beside a column can appear to settle the matter. That confidence is premature. A descriptive statistic summarizes the records currently in view. It does not explain how those records came to exist, why they differ, or what happened outside the dataset's ceiling.

This module concerns the numbers the Explorer reports once a view has been built. The previous module showed how to create filtered populations. This module asks how to read the summaries attached to those populations. The governing distinction remains the same: descriptive statistics report patterns; they do not explain causes.

The Explorer reports several summaries. For categorical columns, which contain labels or categories, it reports counts and percentages. For numeric columns, it reports those summaries

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along with the mean, median, and standard deviation. The variables that recur in this module include time, the presumptive sentence duration in months; moc1, moc2, and moc4, successive characters of the Minnesota Offense Code as represented in the Explorer file; sex, the sentenced person's sex as recorded in the dataset; history, the criminal-history score; and tweight, the combined severity weight of additional felony counts at sentencing.

These variables belong to a sentencing system that already has a legal architecture. MSGC explains that presumptive sentences are based on offense severity and criminal-history score, and that a presumptive sentence includes both a presumptive disposition and a presumptive duration (MSGC, n.d.-a, n.d.-b). That legal context matters before any statistical interpretation begins. The number is not floating in space. It was produced by a grid, a conviction offense, a criminal-history calculation, and a sentencing event.

Each statistic is useful. Each can also be misread. The work of this module is to make the useful reading harder to miss.

The Mode: The Most Common Category

For a categorical variable, the simplest summary is the mode. The mode is the most frequent value in the current view. Reading a mode requires three pieces of information: the value itself, the count of rows with that value, and the percentage of the current view represented by that count.

On the full, unfiltered Explorer dataset, the mode of moc1 is D, which functions in this file as the drug-offense branch of the Minnesota Offense Code. It appears in 79,624 of the 294,467 records, or approximately 27.04 percent of the dataset (Minnesota Sentencing Explorer, 2026). That is a precise claim about one variable in one file. It should not be widened into a claim that drug crimes were the most commonly committed crimes in Minnesota during the period. It should not even be

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widened carelessly into a claim about the broadest offense-type variable, because the uploaded file's broader offtype variable has property offenses as its modal category.

The distinction is not pedantic. The dataset records sentencing records that reached the bottom of the criminal-justice funnel. It does not record all conduct, all arrests, all charges, all dismissed cases, or all negotiated pleas. A mode describes the dataset's rows. It does not describe the entire world upstream of those rows.

Counts and Percentages

A count answers "how many." A percentage answers "how common within the group." The two questions often travel together, but they are not the same question.

The difference becomes especially important when groups are unequal in size. A larger group may have a higher raw count for nearly every outcome simply because the group contains more rows. The count may be larger even when the outcome is less common within that group. In that setting, the denominator is not a technical detail. It is the whole interpretive problem.

Sentence values by sex illustrate the point. In the uploaded Explorer file, 241,387 records are coded male, 53,073 are coded female, and seven are coded corporate (Minnesota Sentencing Explorer, 2026). Men's records therefore make up approximately 81.97 percent of the file, while women's records make up approximately 18.02 percent. That imbalance means raw counts will often be larger for men even when a value is proportionally more common among women.

The 12.03-month value shows the trap directly. In the uploaded file, time == 12.03 appears in 52,999 male records and 19,485 female records. The male count is larger. But within each sex category, the value appears in approximately 21.96 percent of male records and 36.71 percent of

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female records (Minnesota Sentencing Explorer, n.d.). Both statements are true. They answer different questions.

The legal context helps explain why the value is recognizable. Minnesota defines a felony as a crime for which a sentence of imprisonment for one year or more may be imposed (Minn. Stat. § 609.02, 2025). The Explorer's 12.03-month value functions as a year-and-a-day marker in the dataset. The proper statistical reading, however, does not depend on treating that value as a moral boundary. It depends on reading the denominator correctly.

This pattern is also consistent with a broader sentencing literature showing that gender differences in sentencing outcomes are empirically important and difficult to interpret without attention to offense mix, criminal history, charging, plea bargaining, and case selection. Mustard (2001) found gender disparities in federal sentencing after controlling for legally relevant factors. Starr (2015) argued that much observed gender disparity in federal criminal cases is generated before final sentencing, including through charging, plea bargaining, and sentencing fact-finding. Those studies should not be imported as estimates for Minnesota. Their value here is methodological: they warn the reader not to explain a sex difference from a descriptive table alone.

Here is a task a student could undertake with the Explorer to understand why percentages are usually better than counts when comparing groups of different sizes. The student should clear the Explorer, summarize sex, and then compare the distribution of the 12.03-month value within each sex category. This task was chosen because men's records make up a much larger share of the dataset, which makes raw counts especially vulnerable to denominator confusion. What the student should notice is that one group can have the larger count while another group has the larger percentage. What the task is meant to teach is that counts answer "how many," while percentages answer "how common within the group."

Two Kinds of Middle: Mean and Median

Numeric variables require a different kind of summary. The mean and the median are both measures of center, but they define the center differently.

The mean sums all values and divides by N. The median orders the values and identifies the middle value, so that half the rows fall below it and half above it. A small example shows the difference.

For the values 2, 3, 4, 5, and 26, the median is 4. The mean is 8. One large value pulls the mean upward, while the median remains at the center of the ordered list.

That small example is the arithmetic structure of much sentencing data. Many records have shorter presumptive durations. A smaller number have much longer presumptive durations. The long values do not need to be common to matter. They pull the mean.

NIST's guidance on exploratory data analysis explains why this matters. Measures of variability and center respond differently to tail behavior, and long tails can distort mean-centered summaries (NIST, n.d.-a). NIST also defines positive skew as a distribution with a long right tail, which is exactly the shape students should learn to recognize in many sentence-duration summaries (NIST, n.d.-b).

A filtered view can make the contrast visible. If the Explorer is filtered to assault using `moc1 == A` and then to terroristic threats using `moc2 == 9`, the resulting view isolates 12,110 records. In that filtered view, the mean of time is 16.13 months, the median is 15 months, and the standard deviation is 5.59 months (Minnesota Sentencing Explorer, n.d.). The mean and median sit close together because the filtered population is more compact than the all-cases view.

Clearing the filters and reading time across the full felony sentencing dataset produces a different shape. In the unfiltered view, among records with nonmissing time, the mean is 27.20 months, the median is 17 months, and the standard deviation is 29.76 months (Minnesota Sentencing Explorer,

2026). The all-cases view combines many offense categories and severity levels. The mean sits above the median because longer presumptive durations pull it upward.

The rule is portable. When the mean is noticeably higher than the median, the reader should ask whether a long right tail is pulling the mean upward. The statistic is not wrong. The interpretation must be disciplined.

Why the Median Often Describes the Typical Case

The median is useful because it is robust. It resists extreme values. If the example above changes from 2, 3, 4, 5, 26 to 2, 3, 4, 5, 6, the median remains 4, while the mean changes substantially. One value moved. Only one measure of center moved with it.

That property matters in felony sentencing records because very long presumptive durations are not errors to be cleaned away. They are part of the system being described. MSGC's grids are built around offense severity and criminal-history score, and some offense categories carry much longer presumptive durations than others (MSGC, n.d.-b). Their presence makes the mean important, but it also makes the median especially useful when the question concerns the typical record.

A claim that the "average sentence" is 40 months should therefore produce an immediate question: mean or median? If the claim uses the mean, it may describe the arithmetic burden of the distribution. If it uses the median, it may describe the middle case more directly. Both can be appropriate. Neither should be left unnamed.

The mean still matters. Because the mean multiplied by N equals the total, it is useful when the question concerns aggregate burden. A corrections agency projecting prison capacity or a budget office estimating system costs may care about total months, not the typical record. Minnesota law explicitly directs the Commission to consider correctional resources and to conduct ongoing

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research on imprisonment and alternatives to imprisonment, which makes aggregate burden a legitimate system-level concern (Minn. Stat. § 244.09, 2025). Those are mean questions. A reporter or student describing what happens in the typical case is usually asking a median question.

The lesson is not “median good, mean bad.” The lesson is that each statistic answers a different question. The disciplined reader matches the statistic to the claim.

Spread: The Standard Deviation

Two views can share a mean and still describe very different distributions. One may cluster tightly around the mean. Another may scatter widely across a long range. The standard deviation helps describe that difference.

The standard deviation measures spread in the original units of the data. NIST defines it as the square root of the variance and explains that it restores spread to the original measurement units, unlike variance, which squares those units (NIST, n.d.-a). For time, this means the standard deviation is read in months.

A small standard deviation indicates that values are clustered closely. A large standard deviation indicates that values are widely dispersed. No formula is necessary at this stage. The important skill is comparative reading.

Across the full Explorer dataset, the standard deviation of time is 29.76 months, slightly larger than the mean of 27.20 months among records with nonmissing time (Minnesota Sentencing Explorer, 2026). That pairing tells the reader that the distribution is widely spread. It also warns against treating the mean as a strong description of any individual record.

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Within the terroristic-threats filtered view, the standard deviation of time is 5.59 months (Minnesota Sentencing Explorer, n.d.). The statistic has not changed its meaning. The population has changed. The narrower view contains records that are more alike in presumptive duration.

Two readings follow. When the standard deviation of time is nearly as large as, or larger than, the mean, the values are widely scattered and the mean may be a poor description of any individual record. When the standard deviation is close to zero, the records are tightly clustered and the mean is more representative of the cases in view.

The visual lesson matters as much as the numeric one. NIST emphasizes that histograms are effective for showing spread, tail behavior, skewness, and kurtosis (NIST, n.d.-a, n.d.-b). A table can report the standard deviation. A histogram can show why the standard deviation looks that way.

Where Spread Comes From: Variance as Composition

The deepest lesson in this module is that spread often comes from mixture. A wide distribution may reflect a chaotic process, but it may also reflect a view that combines unlike cases.

The variable tweight illustrates the point. In the unfiltered Explorer dataset, tweight has a mean of 1.70, a median of 1.00, and a standard deviation of 2.35 (Minnesota Sentencing Explorer, 2026). Those numbers describe a mixed population. Some sentencing records have little or no additional felony weight. Others carry much more.

Filtering to history == 0 narrows the view to records for sentenced people with a criminal-history score of zero. In that filtered population, the standard deviation of tweight drops to 0.15 (Minnesota Sentencing Explorer, 2026). Nothing about the formula for standard deviation has changed. The composition of the view has changed.

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That change matters because it teaches a method. Filtering is more than a way to search. It is a way to ask how much of a pattern remains when part of the population's composition is held constant. If a statistic changes dramatically after a filter is applied, the change is not a nuisance. It is information about the relationship between the statistic and the population being summarized.

A compact reading brings the pieces together. The unfiltered time summary reports a mean of 27.20 months, a median of 17 months, and a standard deviation of 29.76 months. The proper interpretation is descriptive: among records with nonmissing time in the current view, the typical record sits closer to 17 months, the mean is pulled upward by longer presumptive durations, and the values are widely dispersed. The pattern is visible. The cause is not yet established.

The Final Rule: Check N First

Every statistic depends on the number of rows beneath it. The Explorer will compute a mean, a median, and a standard deviation even when the view contains only a handful of records. The output may look official. The denominator may be too small to support much interpretation.

The point is clearest in an extreme case. If the Explorer is filtered to weapons offenses using `moc1 == W` and then to `moc4 == 8`, the resulting view contains only four records in the uploaded file. The Explorer can still summarize time. With $N = 4$, however, those summaries should be read as four case records, not as a stable pattern.

This is not only a statistical caution. It is an ethical one. A small N can turn a table into a disguise. It gives anecdotes the costume of analysis.

The rule should become automatic: check N first. Before trusting a mean, median, percentage, mode, or standard deviation, identify how many rows the statistic describes. Precision on the screen is not the same as strength in the evidence.

Summary

The core lesson of this module is that descriptive statistics summarize the current view, and every current view has a population, a denominator, and a limit. The mode identifies the most common category. Counts answer how many. Percentages answer how common within a group. The median describes the middle record, especially when the distribution is skewed. The mean describes arithmetic burden and is sensitive to extreme values. The standard deviation describes spread, which often reflects mixture rather than disorder.

The mistake this module prevents is letting a statistic carry more meaning than the dataset can support. From this point forward, every descriptive number should be read in order: name the population, name the variable, read the statistic, interpret it descriptively, and state the limitation. The next module can then ask a more difficult question: when a pattern appears, how much of it remains after the population is narrowed?

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Unit II: The Machine — Minnesota's Sentencing System

Module 4: Classification, Felonies, and the Offense Code

Classification, Felonies, and the Offense Code: How the System Decides What a Case Is

Felony or misdemeanor? Assault in the first degree or terroristic threats? Criminal justice and law lean heavily on the categorization of human conduct, an activity that is, on close inspection, messy, contested, and rarely as clean as the finished label suggests. The previous unit established

a discipline for reading numbers: locate them in the funnel, identify the unit of analysis, and keep description separate from explanation. This chapter turns that discipline on the categories themselves. Before a sentence can be counted, the conduct behind it must be sorted into a class the system recognizes. The Minnesota Offense Code (MOC) that labels each record in the dataset is the last of those sortings. Understanding what the code does, and what it does not, is the aim of this module.

The organizing question is simple: **why do we classify crimes at all?** The answer is that classification is not one act but a chain of them, performed by different actors at successive stages of the criminal-justice funnel, each for a different reason. Following that chain from the top of the funnel to the bottom reveals both the utility of classification and its central limitation, the limitation that governs how the dataset must be read.

Classification Along the Funnel

At the entry point: the police. A crime enters the funnel when police classify conduct as a crime in the first place (*or fails to enter when they do not*), and the felony/misdemeanor line shapes what they may do about it. That line is not merely descriptive; it carries legal force. Under long-settled Fourth Amendment doctrine, an officer may make a warrantless arrest for a felony on probable cause whether or not the offense occurred in the officer's presence, whereas warrantless arrests for misdemeanors are more tightly constrained: classically, by the requirement that the offense be committed in the officer's presence (LaFave, 2020). Classification, in other words, allocates police authority before it does anything else.

It also allocates attention. Police work is defined less by mechanical rule-following than by discretion, and departments differ systematically in how they exercise it, Wilson's (1968) classic

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study distinguished "watchman," "legalistic," and "service" styles of policing, each reflecting a different local judgment about which offenses warrant intervention and which do not. The felony line functions here as an administrative threshold: it marks the level of seriousness at which the system commits its more intensive machinery. This is why the felony is, in the draft's phrase, a useful line in the sand. The alternative would be recording and analyzing every speeding ticket and petty theft with the full apparatus reserved for serious crime, a completely untenable situation. Misdemeanors are not trivial in aggregate; they constitute roughly 80 percent of the nation's criminal dockets, some thirteen million cases a year (Natapoff, 2018). But they are handled on a separate, higher-volume, lower-intensity track. The dataset students explore contains none of them. It records only felonies, which are conduct the law designates as most serious, carrying a potential term of more than a year. The Minnesota Sentencing Commission does so precisely because the felony threshold is where the guidelines system trains its attention.

At the charging stage: prosecutors. Once conduct is reported and an arrest made, the prosecutor translates events into a formal accusation. A charging decision is tied to probable cause to believe the accused violated a *specific statute* (Barnum, 2024), and statutes are themselves acts of categorization: the legislature has already sorted conduct into felony, gross misdemeanor, and misdemeanor bins. But the prosecutor's selection among available charges is consequential in its own right. Discretion at this stage is broad, and scholars have argued that the charge a prosecutor selects "almost predetermines the outcome of a criminal case," since overcharging and charge bargaining shape which statutory category is ultimately applied (Davis, 1998). This matters for reading the data. As the counting rules from Unit I established, a record is described by its *most serious* offense; lesser charges are absorbed. Charges that drop away through bargaining or merger are a form of attrition *within* categories. This is the next place where events are selected out

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of the funnel. Put another way this is the same selection logic as the dark figure of crime, those events which never interact with the system at all, only operating one level down. What survives to be coded is not the whole of what happened but the top charge the system chose to pursue.

At the sentencing stage: judges. When a judge imposes a sentence, offense classification becomes a direct input to the outcome. The intellectual foundation is the principle of proportionality. The system intends that the severity of punishment should be proportionate to the seriousness of the offense, this is articulated influentially by von Hirsch (1976), whose "just deserts" framework shaped the development of sentencing guidelines in the United States and, particularly, in Minnesota. Minnesota's guidelines operationalize that principle as a grid whose two axes are offense severity and criminal history. Offense classification is thus one of the grid's two coordinates; it is half of what determines the presumptive sentence recorded in the 'time' column. The mechanics of that grid are the subject of the next module. For now, the point is that classification does not merely describe the case to the judge, it partly *determines* the recommended punishment.

Below and beyond the funnel: the public and the state. A classification does not expire when a sentence is served. Long after the funnel has done its work, the label persists and continues to govern access to ordinary life, a body of restrictions scholars call the *collateral consequences* of conviction. Chin (2012) describes the aggregate effect as a kind of "civil death": tens of thousands of federal, state, and local provisions restrict the employment, benefits, residence, and civil status of people with convictions, most of whom were never imprisoned at all. Two examples show how finely these consequences track the *type* of felony.

The first is public benefits. A 1996 federal law (the Personal Responsibility and Work Opportunity Reconciliation Act, § 115, codified at 21 U.S.C. § 862a) imposed a lifetime ban on TANF and SNAP,

cash assistance and food benefits, for people convicted of a *drug* felony, while permitting states to opt out of or modify the ban. The provision's logic is worth pausing on, because it inverts what one might expect: it applies only to drug felonies and to *no other felony*, however serious. A person convicted of a violent felony, up to and including murder, faces no comparable federal benefits bar; a person convicted of felony drug possession may face a lifetime one. States have largely softened the rule, by late 2023 only one state, South Carolina, had neither opted out of nor modified either ban (Love & Sibilla, 2023), but the example illustrates how a single classificatory line, drawn decades ago for reasons of drug policy, still allocates access to food and cash assistance today.

The second is employment. Employers commonly screen not merely for the *existence* of a felony record but for its *kind*. Research on employer hiring behavior finds that willingness to hire a person with a record varies systematically by offense type: employers are markedly more averse to applicants with violent convictions than to those with drug or property convictions (Holzer, Raphael, & Stoll, 2006). Classification makes that distinction operable—it renders "what kind of felony" a concrete, checkable fact. (Whether such screening is fair, and what harm it produces, is taken up under **Downsides**, below.)

At the far end: scholars. Finally, classification is what makes the systematic study of crime possible at all. It does so in two distinct ways. First, standardization: shared offense definitions let researchers aggregate and compare data across thousands of jurisdictions and across decades, the achievement that national systems such as the FBI's Uniform Crime Reporting program and its successor, the National Incident-Based Reporting System, were built to deliver (Mosher, Miethe, & Hart, 2011). Without common categories, one county's "assault" could not be compared to another's, and no national picture could be assembled. Second, theory-building: offense

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typologies let scholars generalize beyond individual cases and accumulate knowledge, treating classes of offense as objects of explanation. The construction of crime types is, in this tradition, a precondition for cumulative criminological theory rather than an afterthought to it (Clinard & Quinney, 1973). Every finding in this resource rests on that foundation: the analyses are possible only because the conduct behind 294,000 sentences was sorted into comparable categories in the first place.

The Thesis: The MOC Records a Decision, Not an Event

Trace the chain and its lesson emerges. Police classify conduct as crime; prosecutors translate conduct into a charge; judges translate the charge into a guideline category; and the MOC records the system's final internal label. At each step the category is *re-made*, refined, narrowed, sometimes swapped for another, by human decision. The Minnesota Offense Code is a system used in Minnesota to classify offenses internally, at the level of the state's own recordkeeping. It is not a neutral description of what a person did; it is the residue of a chain of decisions about what the system determined the case to *be*.

This is Sellin's Dictum from Unit I, applied now to categories rather than to counts. Just as felony sentencing data is an excellent record of what the system *decided* and a poor proxy for how much crime *occurred*, the MOC is an excellent record of the system's final classification and a poor proxy for the underlying conduct. A record coded to a given offense reflects the charge that survived every prior stage, not the full reality of the event. Reading the code as a transparent window onto behavior is the categorical version of the same error Unit I warned against. This beckons further discussion, but entire entries are only representative of the *highest*, most severe count, and in a case where someone murders one person and nearly kills another only the murder would appear as an entry, thus the dataset is not indicative of the question "How many people were attacked?"

The Four Families

Beneath the fine grain of the MOC lies a coarse partition that criminologists use to organize almost any body of offense data. Conventionally, offenses are grouped into four "families": **person** (harm directed at a body), **property** (harm to ownership or possession), **drug** (offenses defined by controlled substances), and **public order** (offenses against collective norms or the functioning of institutions). This four-way scheme is the working vocabulary of much criminal-justice statistics. The Bureau of Justice Statistics, for instance, routinely reports violent, property, drug, and public-order offenses as distinct groups. It maps closely, though not exactly, onto the official federal taxonomy: NIBRS sorts every offense into **Crimes Against Persons**, **Crimes Against Property**, and **Crimes Against Society**, folding both drug and public-order offenses into that third "Society" category (Federal Bureau of Investigation, n.d.). The module's four families simply split "Crimes Against Society" back into its drug and public-order halves.

That the official scheme has three categories and the analytic one has four is a first clue to a deeper point: **categorization is an act of human judgment, not a reading of nature**. Boundary cases make this vivid. Robbery, theft accomplished by force, sits astride the line between the person and property families, and different systems resolve it differently: it is a *person* crime under the FBI's older Summary Reporting System but a *property* crime under NIBRS. Nothing about robbery changed; the classifiers made a different choice. Such edge cases are features, not flaws. They reveal the work a classification is doing.

The most influential account of that work comes from outside criminology. In **Sorting Things Out**, Bowker and Star (1999) argue that "to classify is human" (p. 1), that every classification system, however technical it appears, encodes choices and values, advantaging some perspectives while silencing others, and that this constructedness is usually rendered invisible by the finished

categories themselves. Their lesson reprises the discipline from Module 1, where every column was to be treated as an administrative decision first and a measurement second. The MOC is exactly such a decision. It was built to run Minnesota's sentencing system, and its categories carry the priorities of that system rather than the neutral contours of human conduct.

Working with the MOC in the Explorer, one navigates this classification by *drilling down* one character at a time. The first character, ``moc1``, names the top-level family; successive characters narrow the offense within it. The exercises below explore one top-level code exemplifying each family, to see how each type of offense is classified and, more instructively, to see where the categories surprise us.

Explorer Task A. Read the value counts for ``moc1`` on the full dataset. Note how large a share the person-crime categories that dominate public attention: homicide & assault, actually occupy among *felonies*, the most serious offenses of all.

Explorer Task B. Drill into assault (``moc1 == A``) and read the sub-code distribution. Observe what actually dominates the category, whether physical assault, or the verbal-threat offenses one might not expect to find at the top of an "assault" bin.

The second task previews a recurring theme: what a category *contains* is often not what its name implies. That gap between label and content is not a defect in the data but a fact about how classification and charging work and it is one reason media impressions of crime mislead.

Why the small-share finding is surprising. Homicide and violent assault command a share of public attention wildly out of proportion to their frequency, because news coverage overrepresents them. A content analysis of a full year of local crime reporting in Milwaukee found that homicide accounted for a majority of the front-page crime stories in the daily newspaper and a large share

online, even though homicides are so rare relative to total crime as to be nearly invisible in the underlying police statistics; property offenses, conversely, made up the great bulk of actual crime but a small fraction of coverage (O'Hear, 2020). The pattern is long-standing. Decades earlier, researchers found that the public's picture of crime tracked newspaper portrayals far more closely than official crime counts (Sheley & Ashkins, 1981). This is why discovering the modest share of violent offenses even among felonies can feel counterintuitive: the data corrects an impression the media has built. What the dataset's modal categories reveal is not what crime *feels* like but what the criminal legal system actually processes.

Downsides to Classification

These classifications are indispensable, as the walk down the funnel showed. They are also open to serious criticism.

The first criticism concerns *coherence*. Classification is meant to track relative seriousness. In premise it aims to sort conduct so that graver wrongs sit in graver categories. Scholars have shown that the felony/gross-misdemeanor/misdemeanor grading of statutes often drifts from that ideal. In an empirical study, a research team surveyed a representative sample of Pennsylvanians about the seriousness of many offenses and compared public judgment to the statutory grades those offenses actually carried, finding frequent and substantial mismatches (Robinson, Gaeta, Majarian, Schultz, & Weck, 2010). In one striking instance, possessing wiretapping equipment was rated by the public as roughly comparable to stealing goods worth \$50 to \$200, an offense whose grade implies a maximum of about two years, yet the wiretapping offense is graded as a felony carrying up to seven years. The difference is not merely academic; the grade determines exposure to years of imprisonment. If grading can diverge this far from shared intuitions of seriousness, then

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the categories students analyze are administrative artifacts to be read critically, not natural measures of harm.

The second criticism concerns *consequences*. The walk down the funnel noted that classifications outlive the sentence, governing benefits and employment. Those uses carry real benefits for employers and agencies, but they also inflict real harm. The foundational evidence comes from an audit study in which matched applicants, differing only in whether they reported a criminal record, applied for entry-level jobs: a felony record roughly halved the callback rate for white applicants and cut it by nearly two-thirds for Black applicants (Pager, 2003). It is also worth considering however that later field experiments confirm the record's weight but complicate the remedies: when employers were barred from asking about records at all, some appeared to substitute racial assumptions, widening the very gap the policy sought to close (Agan & Starr, 2018). Classification, then, is double-edged. The same categorical precision that lets the state monitor its own sentencing also lets a label follow a person for a lifetime.

Summary

Classification is the machinery that turns human conduct into a countable record, and it operates at every stage of the funnel. Police use the felony line to allocate authority and attention and to set the threshold for intensive recordkeeping; prosecutors translate conduct into charges, selecting among categories in ways that shape the outcome; judges use offense severity as one of the two axes that fix the presumptive sentence; the public and the state use the surviving label to govern benefits and employment long after the case closes; and scholars depend on standardized categories both to compare data and to build theory. The Minnesota Offense Code sits at the end of this chain. It records what the system decided a case **was**, not what happened—Sellin's dictum in categorical form. Beneath its fine grain lie four families—person, property, drug, and public

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order—whose boundaries are acts of human judgment rather than facts of nature, as edge cases like robbery and the mismatches of statutory grading both reveal. Read this way, the MOC is neither a neutral mirror of crime nor a flaw to be corrected, but an administrative decision to be interpreted with care. The next module opens the grid on which offense severity and criminal history combine to produce the number in the `time` column.

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Module 5: The Grid or How a Sentence Becomes a Number

Introduction

A felony sentence can look like an individualized judgment. A judge stands in a courtroom, the defendant stands before the court, and a sentence is pronounced. That scene can make the number appear personal, as though it came directly from the facts of the case and the discretion of the court.

That reading is too hasty. In Minnesota felony sentencing, the number is often produced by a legal machine: the sentencing grid. The grid does not eliminate judgment, but it structures it. It converts two inputs, offense severity and criminal-history score, into a presumptive sentence. Once those inputs are known, the expected sentence is no longer a mystery. It is a cell.

This module explains how Minnesota's sentencing grid turns a conviction into a number of months. It also explains why the severity column in the dataset is more complicated than it first appears. The field looks like one variable. It is actually four coding systems stacked into one column: the Standard Grid, the Sex Offender Grid, the Drug Offender Grid, and a small set of off-grid first-degree murder records. The mistake is understandable. The correction is essential.

The Promise of the Grid

Before 1980, Minnesota felony sentencing gave judges broad discretion, while the parole board played a substantial role in determining how long a person actually remained in prison. Two people convicted of the same offense, with similar records, could leave court with very different futures. Reformers identified that problem as sentencing disparity: a measured difference in sentencing outcomes among similarly situated cases. By the late 1970s, disparity had become one of the central critiques of American sentencing (Frase, 2005, pp. 132, 139, 142).

Minnesota's response became nationally significant. In 1978, the Legislature created the Minnesota Sentencing Guidelines Commission, and on May 1, 1980, Minnesota became the first state to implement presumptive sentencing guidelines (Frase, 2005, pp. 131, 143, 150; Minn. Stat. § 244.09, subds. 1, 12, 2025). The word presumptive matters. These were not merely advisory suggestions for judges to consult. The legal structure was more precise: although Minnesota law describes the Sentencing Guidelines as “advisory,” the statute also requires the district court to follow guideline procedures, and the Guidelines establish presumptive fixed sentences keyed to offense and offender characteristics (Minn. Stat. § 244.09, subd. 5, 2025). Within that framework, the presumptive sentence is treated as the proper sentence for the ordinary case: the Guidelines deem presumptive sentences appropriate for covered felonies and permit departures only when substantial and compelling circumstances can be identified and articulated (Minnesota Sentencing Guidelines Commission [MSGC], 2024, § 1.A.6). The rule-like character of the system becomes clearest at the point of departure: the court must pronounce the applicable disposition and a sentence within the applicable prison range unless identifiable, substantial, and compelling circumstances support a departure, and any deviation requires written findings explaining the departure (Minn. Stat. § 244.10, subd. 2, 2025; MSGC, 2024, § 2.D.1). Frase describes the same

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structure historically: the Guidelines specified the sentence “presumed to be correct” for each offense-severity and criminal-history combination, and judges could depart only by citing substantial and compelling circumstances (Frase, 2005, p. 138).

The statutory purpose was explicit. Minnesota law directs the Commission to establish guidelines that are “neutral with respect to the race, gender, social, or economic status of convicted felons” (Minn. Stat. § 244.09, subd. 5). The grid is the institutional mechanism built to serve that promise. It translates a felony conviction and a criminal-history score into a presumptive sentencing outcome.

That promise should be read carefully. The grid does not make sentencing simple. It does not remove all discretion from the system. It does not explain every sentencing outcome in the dataset. What it does is structure the typical case. For the records in this dataset, the grid is the central machine through which presumptive sentence duration becomes visible.

The Machine Itself

Two axes dominate the differentiation between sentences in the grid system (data).

The first axis is offense severity. Every felony offense covered by the guidelines is assigned a severity level by the Minnesota Sentencing Guidelines Commission. The severity level attaches to the conviction offense, meaning the offense of conviction, not necessarily the original charge filed at the beginning of the case (Minnesota Sentencing Guidelines Commission [MSGC], 2019, § 2.A.1). That distinction matters because the dataset sits downstream from charging, plea bargaining, dismissal, amendment, conviction, and sentencing. A severity level records where the case landed at sentencing. It does not reconstruct every earlier stage.

The second axis is criminal-history score. Criminal-history score is a point total built primarily from prior felony convictions, with the grid columns running from zero to six or more (MSGC, 2019, §

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4.A). From the perspective of the dataset, once the severity row and criminal-history column are known, in most cases the standard distribution of *time* collapses drastically (data).

Each cell answers two questions at once. The first is disposition: whether the sentence is presumed to be executed or stayed. Under the Minnesota Sentencing Guidelines, a presumptive disposition is the recommendation for either commitment or a stayed sentence; cells outside the shaded area of the Minnesota's grids are presumptive commitments, while cells within the shaded area are presumptive stayed sentences (Minnesota Sentencing Guidelines Commission [MSGC], 2024, § 1.B.13). A stayed sentence may take the form of a stay of execution, in which the prison sentence is pronounced but its execution is stayed, or a stay of imposition, in which imposition of the sentence is stayed to a later date (MSGC, 2024, § 3.A.1). When a court stays imposition or execution, Minnesota law permits the court to place the defendant on probation, with or without supervision, and on conditions prescribed by the court (Minn. Stat. § 609.135, subd. 1).

The second question is duration: how many months the sentence carries. For prison cells, the grid supplies a fixed presumptive duration and a permissible range around that duration. Minnesota law allows a judge to sentence within a range from 15 percent below to 20 percent above the fixed duration without formally departing (Minn. Stat. § 244.09, subd. 5(2)). The asymmetry matters. The authorized range is wider upward than downward.

A judge who wishes to go outside the presumptive sentence must identify “substantial and compelling circumstances” and state the reasons in writing (Minn. Stat. § 244.10, subd. 2). Departures are therefore not invisible, casual adjustments. They are exceptions that require justification. For the typical case, the grid cell is the sentence.

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The portable rule is this: time is the output of a table lookup. That fact predicts much of the shape of the dataset. It also explains why the severity field must be read before it is summarized.

Here is a task a student could undertake with the Explorer to understand the lookup structure. The student should compare two hypothetical cases with the same offense severity but different criminal-history scores. This task was chosen because it isolates one axis of the grid while holding the other constant. What the student should notice is that the two cases remain on the same severity row but move into different criminal-history columns. What the task is meant to teach is that the presumptive sentence belongs to the cell, not to the offense alone.

One Column, Four Languages

If *time* is the output of the lookup table, then offense severity is half of the lookup key. It is one of the most important fields in the dataset. It is also one of the easiest fields to misread.

At first glance, the severity column looks broken. It contains numbers from 1 to 11. It contains single letters from A through H. It contains two-digit codes in the fifties. It also contains the literal text “Murder 1” for 354 records.

The column is not broken. It is living historical record.

Administrative data often records the outputs of systems that changed over time without translating older records into a single modern scheme (Rothbard, 2013, p. 19). The severity column is not one clean severity scale. It is four coding regimes stored in one field. Each regime corresponds to a real institutional structure in Minnesota sentencing.

The first habit is to pause when one column contains structurally different kinds of values.

Numbers, letters, and text strings should not be averaged or ranked until the reader knows what system produced them. Rothbard (2013) makes the same methodological point in data-quality

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terms: administrative-data users must inspect frequency distributions, clarify variable definitions, and address changes in coding schemes before treating values as comparable across time or systems (pp. 6, 11–12, 18–19).

Here is a task a student could undertake with the Explorer to understand the four-family structure of the severity column. The student should clear all filters and view the distribution of the severity field. This task was chosen because the pattern is visible before any statistical model is needed. What the student should notice is that the column contains four distinct families: integers from 1 through 11, letters from A through H, fifty-series codes from 51 through 59, and “Murder 1.” What the task is meant to teach is that a variable can look singular while carrying multiple administrative histories.

Family 1: The Standard Grid

The integers belong to the Standard Grid. These are severity levels for felonies other than the sex and drug offenses that later received separate grids. In the modern standard grid, severity levels run from 1 at the low end to 11 at the high end (MSGC, 2019, §§ 1.B, 4.A, 5.A). Level 11 includes the most serious standard-grid offenses, such as intentional second-degree murder. Lower levels include less severe felony categories.

The important historical complication is that the grid did not always have eleven levels. The original 1980 grid had ten severity levels (MSGC, 1980, p. 38). Effective August 1, 2002, the Commission inserted a new Severity Level 7. The old levels 7 through 10 became levels 8 through 11 (MSGC, 2025, p. 9).

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That change makes the same number unstable across time. A record coded severity level 7 before the 2002 change and a record coded severity level 7 after the change may not refer to the same rung of the same ladder. The number is identical. The codebook changed underneath it.

This is a classification lesson, not a technical nuisance. A code is meaningful only in relation to the system that produced it. When the legal scale changes, the value in the column inherits that history. The dataset does not automatically warn the reader that the meaning of level 7 changed in 2002. The reader must know enough about the institution to ask.

The proper reading is therefore cautious. A severity number identifies a standard-grid level, but the date of the offense matters for interpreting how that level fits into the relevant edition of the grid.

Family 2: The Sex Offender Grid

The letters belong to the Sex Offender Grid. Effective for offenses committed on or after August 1, 2006, Minnesota removed criminal sexual conduct and related offenses from the Standard Grid and placed them on a separate grid (MSGC, 2019, § 4.B; MSGC, 2006). To distinguish the new grid from the numeric Standard Grid, the Sex Offender Grid used letters rather than numbers. During the 2001 to 2019 window covered by this dataset, the sex grid ran from A through H.

The letters are not grades in the everyday sense. They are severity levels within a separate sentencing structure. Level A is the highest level, including first-degree criminal sexual conduct. Level H is the lowest level, including failure to register as a predatory offender. That distinction explains the distribution. (MSGC, 2006). Level H dominates the lettered codes because failure to register as a predatory offender was ranked at Severity Level H on the Sex Offender Grid for offenses committed on or after August 1, 2006, and failure-to-register offenses generated

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substantial sentencing volume during the relevant period (MSGC, 2014, pp. 4–5). In the dataset, H appears 4,770 times.

The trap is the letter D. The severity column includes 2,501 records coded with the bare letter “D” (data). A reader familiar with Minnesota’s Drug Offender Grid might be tempted to treat these as drug cases, because the published Guidelines label Drug Offender Grid severity levels as D1 through D9 (MSGC, 2024, § 1.B.17). That would be wrong. A bare “D” is Sex Offender Grid level D (codebook). Drug-grid codes are not stored as bare letters in this dataset.

The mistake is understandable because classification systems do not coordinate their symbols for the analyst’s convenience (Rothbard, 2013, pp. 31–32). Administrative-data interpretation requires attention to the source, documentation, and meaning of each variable, especially when similar values may not represent the same information or when coding schemes change across systems or over time (Rothbard, 2013, pp. 11–12, 18–19). In Minnesota sentencing, that problem is visible in the Guidelines themselves: sex offenses are ranked on the Sex Offender Grid using lettered severity levels, while drug offenses are ranked on the Drug Offender Grid using D1 through D9 (MSGC, 2024, § 1.B.17). The same letter can therefore point to different administrative languages. The proper reading depends on the code family (Rothbard, 2013, p. 19). A bare letter belongs to the Sex Offender Grid. A fifty-series number belongs to the Drug Offender Grid.

A note about timing is also necessary. The Sex Offender Grid later expanded from eight severity levels to nine, changing the relevant range from A through H to A through I (MSGC, 2021, pp. 14–15). That change occurred after the dataset’s 2001–2019 observation window. For this dataset, the relevant letter range is A through H.

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Here is a task a student could undertake with the Explorer to understand the “D” trap. The student should filter the dataset to severity = D and examine the resulting offense family. This task was chosen because it makes a plausible mistake visible before it can contaminate analysis. What the student should notice is that a bare D does not identify drug offenses. What the task is meant to teach is that symbols do not carry meaning by themselves; they carry meaning inside a coding regime.

Family 3: The Drug Offender Grid

The fifty-series codes belong to the Drug Offender Grid. For decades, drug felonies were sentenced on the Standard Grid alongside offenses such as burglary and assault. The 2016 Drug Sentencing Reform Act changed that structure. For offenses committed on or after August 1, 2016, controlled-substance crimes moved onto a separate Drug Offender Grid with nine severity levels, labeled D1 through D9 in the published guidelines (MSGC, 2019, § 4.C).

The reform was not merely cosmetic. It lowered some drug offense severities and reflected a policy choice to reserve prison resources for more serious drug cases. Fifth-degree controlled substance crime, including simple possession under Minn. Stat. § 152.025, sits at D2. Fourth-degree controlled substance crime sits at D4. The grid therefore reorganized how drug convictions entered the sentencing machine.

The dataset, however, does not store the published labels D1 through D9. Instead, it stores codes 51 through 59. The natural hypothesis is that the dataset uses an internal encoding: 51 means D1, 52 means D2, and so on through 59 meaning D9.

A hypothesis about a coding system should be tested, not assumed. This one can be tested by triangulating three kinds of evidence: the legal structure, the timing of the codes, and the

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composition of the records. The legal structure supplies nine drug levels. The dataset supplies nine fifty-series codes. The timing aligns with the August 1, 2016 effective date. The offense-type composition also aligns: cross-tabulation shows that 99.97 percent of 5X-coded records are drug offense records. Finally, the frequency pattern supports the interpretation. Code 52 accounts for 8,711 of the roughly 12,000 fifty-series records, about 72 percent of the family. In the post-DSRA drug grid, D2 corresponds to felony fifth-degree controlled-substance crime, including qualifying fifth-degree possession under Minn. Stat. § 152.025, subd. 2(1) and subd. 4(b) (Minnesota Sentencing Guidelines Commission [MSGC], 2024, § 5.B; Minn. Stat. § 152.025, subds. 2, 4, 2025), historically this is the state's highest-volume drug felony (MSGC, 2013, p. 5).

No public codebook needs to announce the answer for the pattern to become legible. The meaning is recovered by reading the law, the timeline, and the data together. When documentation runs out, structure becomes evidence.

That conclusion should still be phrased carefully. The dataset supports the interpretation that 51 through 59 encode Drug Offender Grid levels D1 through D9. The strongest evidence is not merely that there are nine codes and nine levels. A count match is suggestive, but weak on its own. The stronger evidence is that the codes appear in the right period, attach overwhelmingly to drug offenses, and reproduce the expected frequency signature of Minnesota drug sentencing after reform.

Here is a task a student could undertake with the Explorer to understand structure as evidence. The student should filter to the fifty-series severity codes and examine offense type and sentencing year. This task was chosen because the public-facing labels do not appear directly in the field. What the student should notice is that the fifty-series codes are almost entirely drug records and appear only after the drug grid took effect. What the task is meant to teach is that undocumented

codes can sometimes be interpreted by testing whether legal structure, timing, and empirical composition all point in the same direction.

Family 4: The Case the Grid Refuses to Score

The final severity family is not a number and not a letter. It is the text string “Murder 1.” There are 354 such records in the dataset.

First-degree murder is different because Minnesota law imposes a mandatory life sentence (Minn. Stat. § 609.185). There is no grid lookup to perform. The Minnesota Sentencing Guidelines state that no severity level is assigned to first-degree murder because the punishment is mandatory life by statute (MSGC, 2019, § 2.A.3 cmt.). Some of the most serious sex offenses are excluded in a similar way (Minn. Stat. § 609.3455, subd. 2; MSGC, 2019, § 4.B).

The text string is therefore not sloppy data entry. It is the severity field recording that the case did not enter the grid. A grid produces months. A mandatory life sentence is not a grid cell with a monthly duration. The field refuses to score the case because the legal system refuses to send it through that part of the machine.

This produces an important asymmetry. Second-degree murder sits on the Standard Grid at level 11. First-degree murder sits outside the grid entirely. The most serious homicide category is not the top row of the grid. It is off-grid. The grid governs the broad middle of felony sentencing; the statutory extremes are handled directly.

The practical consequence is methodological. The *time* field for these 354 records is not a usable duration. Some records are blank, while others store large placeholder values. If these records are left in a calculation of average sentence duration, mandatory-life cases can silently distort the

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mean. The careful analyst excludes or separately handles off-grid mandatory-life records before summarizing time.

The rule is simple: “Murder 1” is not level 12. It is the outside of the grid.

Why the Mess Exists

Once the four families are placed on a timeline, the severity column stops looking chaotic.

From 2001 through the 2002 change, the sentencing structure relied on one numeric grid with ten standard severity levels. Effective August 1, 2002, the Standard Grid expanded to eleven levels, and the upper levels were renumbered. Effective August 1, 2006, sex offenses moved to their own lettered grid. Effective August 1, 2016, drug offenses moved to their own grid, stored in this dataset as codes 51 through 59. Throughout the full period, first-degree murder remained outside the grid because it carried a mandatory life sentence.

The severity column is therefore a geological cross-section of institutional change. New layers were added as the sentencing system reorganized itself. Older layers were not translated into a single harmonized scale. The field contains the history of the institution that produced it.

This has an immediate implication for analysis. Severity level 2 dominates the entire column, appearing in 84,511 records, or 28.7 percent of the dataset. The likely explanation is compositional. Before August 2016, high-volume fifth-degree drug possession cases were located in the low numeric levels of the Standard Grid. After the Drug Sentencing Reform Act took effect, those cases moved into the fifty-series codes. If that interpretation is correct, the annual volume of severity level 2 should fall sharply after 2016, while the fifty-series codes should rise from zero.

This is not a claim about crime suddenly changing. It is a claim about classification changing. The conduct may be similar. The filing cabinet changed.

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Here is a task a student could undertake with the Explorer to understand the migration from the Standard Grid to the Drug Offender Grid. The student should first filter to severity level 2 and examine annual volume by sentencing year. Then the student should clear the filter, filter to the fifty-series codes, and examine annual volume again. This task was chosen because the 2016 reform creates a visible hinge in the data. What the student should notice is that severity level 2 deflates after the reform while the fifty-series appears. What the task is meant to teach is that changes in a distribution may reflect changes in classification rather than changes in underlying conduct.

The Grid Is Visible in the Standard Deviation

The grid also teaches a statistical lesson. A standard deviation is not just noise. It has composition. Some of the spread in a variable comes from genuinely different cases being pooled together. Some comes from variation within like cases. The sentencing grid makes that distinction unusually clear because the law tells the reader how the number is generated.

A lookup table implies a specific pattern. Within a single cell, the presumptive sentence should vary little, because the cell assigns the presumptive value. Between cells, the mean sentence should change substantially, because different cells carry different presumptive durations. In other words, much of the variance in time should live between cells rather than within them.

The unit of analysis matters. A grid cell is not severity alone. It is severity and criminal-history score. Filtering only to severity level 8 still pools all criminal-history columns. Because presumptive duration rises across those columns, a severity-only filter will produce a larger standard deviation. That is not a failed demonstration. It is the wrong unit of analysis.

The proper unit is the cell.

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Two caveats are necessary. First, cell variation is not exactly zero. The statutory range and departures allow real presumptive sentences to scatter around the cell value. Second, pooling records from 2001 through 2019 also pools multiple editions of the guidelines. A small amount of remaining variation may reflect the grid changing over time rather than random judicial inconsistency. Even with those caveats, the core principle holds. Within-cell variation is much smaller than between-cell differences.

Here is a task a student could undertake with the Explorer to understand variance as composition. The student should filter to Standard Grid records, then filter to severity level 8 and criminal-history score 0. The student should record the count, mean, and standard deviation of time. Next, the student should clear the criminal-history filter and apply criminal-history score 3 while keeping severity level 8. Again, the student should record the count, mean, and standard deviation. This task was chosen because the two filtered views sit on the same severity row and differ only by criminal-history score. What the student should notice is that standard deviation within each cell is small relative to the difference between the two cell means. What the task is meant to teach is that sentence variation across the dataset is partly the product of cell composition. The grid assigns similar sentences within a cell and different sentences across cells.

The correct interpretation is not that judges in those cells are unusually consistent people. The correct interpretation is institutional. The grid instructs the system to treat typical cases in the same cell similarly. The variance collapses within the cell because the sentencing rule is doing exactly what it was designed to do.

What the Severity Code Cannot Tell You

The severity code is powerful, but it is not omniscient. It records which grid and which rung a conviction offense occupies. It does not record the full facts of the case. It does not describe the harm in ordinary human terms. It does not tell the reader whether the event was desperate, ambiguous, violent, opportunistic, coerced, or unusually damaging.

Two records at the same severity level may describe cases that victims, defendants, lawyers, or jurors would experience as profoundly different. The classification system treats them as the same for a sentencing purpose. That is what classification does. It decides which differences will matter and which differences will be suppressed.

This is not a flaw unique to Minnesota sentencing data. It is a basic condition of administrative records. A category is a decision, and a code is not the event itself. Official statistics reflect the organizational processes through which institutions identify, classify, and record conduct. The dataset therefore records the bureaucratic address assigned to a case after it has passed through the criminal-justice funnel, not an unmediated account of the underlying event (Kitsuse & Cicourel, 1963, pp. 136–137).

The off-grid records make the point even sharper. The most serious records in the dataset, the first-degree murder cases, do not receive a numeric severity value at all. The machine's silence is information. It tells the reader that the grid has an outside and that some of the highest-stakes cases live there.

The core lesson of this module is that offense severity is not one simple scale. It is a historically layered field that records multiple sentencing grids, statutory exclusions, and reforms over time. The mistake it prevents is treating all severity values as if they belong to one continuous ladder.

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From this point forward, every use of severity must begin by asking which grid produced the code. The next module extends the same discipline to criminal-history score, where the past becomes a number and enters the second axis of the grid.

Checkpoints for Interpretation

A student who can read the severity column should be able to identify four code families. Codes 1 through 11 belong to the Standard Grid. Codes A through H belong to the Sex Offender Grid during this dataset's observation window. Codes 51 through 59 belong to the Drug Offender Grid as stored in this dataset. "Murder 1" does not belong to the top row of the Standard Grid. It identifies first-degree murder, which is off-grid because the sentence is mandatory life.

The student should also be able to avoid the bare-D error. A record coded severity "D" belongs to the Sex Offender Grid, not the Drug Offender Grid. The drug grid's published labels use D1 through D9, but this dataset stores those levels as 51 through 59.

The student should understand why standard deviation collapses within a single grid cell. A cell fixes the presumptive sentence for typical cases within it. Variation therefore lives largely between cells, not within them.

Finally, the student should be able to explain how undocumented codes can be verified. The strongest evidence that the fifty-series codes encode drug-grid levels is not simply that there are nine values. It is that the codes appear after the 2016 reform, attach almost entirely to drug offenses, and follow the expected frequency pattern of Minnesota drug sentencing.

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Module 6: Criminal History — The Second Axis

There needs to be a M5 like section for history, but this is going to require a good deal more research.

Module 7: What time Measures

Presumptive, Pronounced, Executed, and Served Time

Throughout these materials, the variable time has done a great deal of work. Students have filtered by offense type, summarized sentence lengths, compared means and medians, and watched the distribution of time change as the view of the data narrowed. Until now, however, the course has deliberately postponed one question that should eventually unsettle every careful reader: what does *time* actually measure?

The tempting answer is simple: *time* must mean how long a person was sentenced to serve. A row with *time* = 48 appears to say something obvious: this person received 48 months in prison. Implied then, they will serve four years in prison. But that reading compresses several legally distinct ideas into one number. In Minnesota sentencing data, *time* is best understood as presumptive sentence duration: the recommended fixed sentence length, measured in months, associated with the applicable sentencing-guidelines grid cell. It is a guideline recommendation, not a complete record of punishment experienced, because the sentence may be stayed, departed from, or divided between imprisonment and supervised release (MSGC, 2024, pp. 5, 7; Minn. Stat. § 244.101, subds. 1–2, 2025).

That distinction changes how every earlier statistic should be read. A median time of 15 months does not necessarily mean that the typical person in the filtered group spent 15 months in prison. A mean time of 48 months does not necessarily mean that the average person served four years. Between the guideline number and punishment experienced stand several mechanisms: whether the sentence is stayed or executed, probation and its conditions, local confinement, dispositional and durational departures, plea-agreement practices, the division of executed sentences between imprisonment and supervised release, and later correctional decisions. MSGC monitoring data capture the sentence pronounced, departure reasons, and local-confinement terms, but do not capture other probation sanctions or subsequent sanctions for probation violations (MSGC, 2026, pp. 21–22, 34, 99–100).

This module therefore revises the meaning of earlier work without discarding it. The earlier comparisons remain valuable, but their claim must be named precisely. When students compare time, they are comparing presumptive sentence duration. They are not automatically comparing actual prison time, total confinement, or the full experience of punishment.

The Naive Reading

Suppose a student filters the Explorer to a single case and sees time = 48. The first guess is natural: this person spent 48 months in prison. That guess is wrong in several ways.

First, the kernel of truth: time is recorded in months. This should already be familiar from earlier modules, especially the recurring value 12.03, which represents one year and one day. However, time is not necessarily the prison term pronounced by the judge. It represents the presumptive duration associated with the sentencing guidelines. Furthermore, even when the judge pronounces

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and executes a prison sentence, the pronounced number is not usually the number of months served inside a state prison.

The correct answer to “How long in prison if time = 48?” is therefore conditional. If the sentence was stayed, the answer may be zero months in state prison, though the person may still serve local jail time as a condition of probation. If the sentence was executed and followed the ordinary statutory structure, the prison portion would usually be 32 months, with the remaining 16 months served on supervised release. If supervised release was later revoked, or if disciplinary confinement changed the expected path, actual prison time could move closer to the full 48 months (Minn. Stat. § 244.101, 2024; MSGC, 2024).

One number. Several legal realities. The work of this module is to keep them separate.

What “Presumptive” Means

Minnesota’s sentencing guidelines generate a presumptive sentence from two central inputs: the severity level of the conviction offense and the person’s criminal-history score. The sentencing grid combines those two axes. Where offense severity and criminal history intersect, the grid supplies the presumptive sentence (MSGC, 2024a).

A presumptive sentence answers two questions at once. The first is disposition: should the sentence be executed as a state-prison commitment, or should it be stayed? The second is duration: what fixed term of months does the grid recommend? These are related but distinct questions. A case may have a presumptive duration, but that does not mean the person will necessarily serve that duration in prison.

The word “presumptive” matters because the grid is not a machine that strips the court of all discretion. Minnesota law requires the Guidelines to provide a range around the fixed duration. For

presumptive executed prison sentences, the permissible range extends 15 percent below and 20 percent above the fixed duration, and a sentence inside that range is not a departure merely because it differs from the exact grid number (Minn. Stat. § 244.09, 2024; MSGC, 2024a). A 48-month presumptive sentence, for example, does not mean that 44 months is automatically a downward durational departure. It may remain within the presumptive range.

This is the first correction to the earlier use of time: the number is not a natural measurement, like height or temperature. It is the output of a legal structure. Offense severity, criminal history, sentencing policy, statutory rules, and administrative recordkeeping all pass through the grid before appearing as a number in the dataset. The grid is the machinery; *time* is one of its outputs.

The First Exit: Stayed Sentences

The largest reason *time* is not time served is that most Minnesota felony sentences are not executed as state-prison sentences. A sentence can be stayed. When a sentence is stayed, the person does not immediately go to state prison. Instead, the court places the person on probation, often with conditions that may include local confinement, treatment, restitution, supervision, or other sanctions (Minn. Stat. § 609.135, 2024; MSGC, 2024a).

Minnesota law recognizes two main stay types. In a stay of execution, the court pronounces a prison sentence but suspends its execution. The prison term exists, but it is not carried out unless the person violates probation. The sentence is, in effect, held over the case. In a stay of imposition, the court does not pronounce a prison duration at all. If the person successfully completes probation, Minnesota law provides that the conviction may be deemed a misdemeanor for certain purposes (Minn. Stat. § 609.13, 2024; Minn. Stat. § 609.135, 2024; MSGC, 2024a).

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A stayed sentence should not be confused with no punishment. A person may still be placed on probation and may still serve local confinement as a condition of that probation. The Guidelines allow local confinement of up to 364 days as a condition of a stayed sentence (MSGC, 2025). Thus, a person may avoid state prison while still experiencing substantial criminal punishment.

The scale of this distinction is large. In 2024, the actual state-prison rate for Minnesota felony cases was 24 percent, meaning roughly one-quarter of felony cases resulted in an executed prison sentence. At the same time, the total incarceration rate, counting either state prison or local confinement, was 88 percent (MSGC, 2026). Those two facts must be read together. Minnesota confines many people convicted of felonies, but it does not usually do so through executed state-prison sentences for the full guideline duration.

This is the first major interpretive rule of the module: before reading time as prison exposure, ask whether the sentence was executed. If it was stayed, the two-thirds prison calculation discussed below does not describe what happened.

The Second Exit: The Two-Thirds Rule

Even when a sentence is executed, the pronounced sentence is not the same thing as expected prison time. For felony offenses committed on or after August 1, 1993, Minnesota divides an executed sentence into two parts: a term of imprisonment equal to two-thirds of the executed sentence and a supervised release term equal to the remaining one-third (Minn. Stat. § 244.101, 2024).

A 48-month executed sentence therefore ordinarily means 32 months in prison and 16 months on supervised release. A 60-month executed sentence ordinarily means 40 months in prison and 20 months on supervised release. The sentencing court is required to explain this structure when

pronouncing sentence, including the total executed sentence, the term of imprisonment, and the supervised release term (Minn. Stat. § 244.101, 2024; MSGC, 2024a).

The two-thirds rule is not a guarantee that the person will permanently leave prison after two-thirds of the sentence. Supervised release can be revoked, and disciplinary violations may extend confinement. Minnesota law permits the correctional system to impose consequences for violations that can change the amount of time actually spent in custody (Minn. Stat. § 244.05, 2024; Minn. Stat. § 244.101, 2024). Still, the two-thirds rule provides the ordinary translation between an executed sentence and its prison component.

This is the second major interpretive rule of the module: even for executed prison sentences, time overstates ordinary prison time if read as months inside a prison. The ordinary prison term is two-thirds of the executed sentence, not the full sentence.

Departures: When the Court Leaves the Presumptive Path

The next gap between time and punishment comes from departures. A departure occurs when the court imposes a sentence outside the presumptive recommendation of the Guidelines. Minnesota distinguishes between dispositional departures and durational departures (MSGC, 2024a, 2026).

A dispositional departure changes the kind of sentence. If the Guidelines recommend a stayed sentence but the court executes prison, the departure is aggravated. If the Guidelines recommend prison but the court stays the sentence, the departure is mitigated. A durational departure changes the length of the sentence. If the sentence falls more than 15 percent below the fixed duration, it is a mitigated durational departure. If it rises more than 20 percent above the fixed duration, it is an aggravated durational departure (Minn. Stat. § 244.09, 2024; MSGC, 2024).

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The direction of departures matters. In 2024, 41 percent of presumptive-commitment cases received mitigated dispositional departures, meaning the Guidelines recommended prison but the court stayed the sentence. As a share of all felony cases, mitigated dispositional departures were about 16 percent (MSGC, 2026). Both statistics are true, but they answer different questions. The first asks: among cases where the Guidelines recommended prison, how often did the court stay the sentence instead? The second asks: among all felony cases, how common was that outcome?

The difference is a denominator lesson wearing a sentencing costume. Whenever a departure statistic appears, the first question should be: departure rate among what population? A percentage of presumptive-commitment cases and a percentage of all felony cases are not interchangeable. Confusing them turns a real statistic into a carnival mirror.

Departures show why *time* should not be read as a simple prediction. The grid gives a presumptive recommendation, but the court may lawfully depart when substantial and compelling reasons justify doing so. A departure is not an error in the data. It is part of the sentencing system the data records.

Plea Bargaining's Shadow

Departures do not capture all discretion. A departure records movement away from the presumptive sentence attached to the conviction offense. But the conviction offense may itself be the product of earlier discretionary choices, including charging, plea negotiation, charge reduction, dismissal, amendment, or the retention of multiple counts. Those choices occur before the applicable guideline cell is selected (Miethe & Moore, 1987, pp. 38-39; MSGC, 2024).

This matters because a case can be transformed before time is ever calculated. If a charge is reduced through plea bargaining, the case may land in a different offense category or severity level.

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The Guidelines then compute a presumptive sentence for the conviction offense, not for every charge initially filed or every fact alleged. From the perspective of the sentencing grid, the sentence may appear compliant. From the perspective of the larger criminal-justice funnel, important discretion has already occurred upstream.

This extends the classification lesson from Module 4. The Minnesota Offense Code records the offense classification that survived the process, not the full underlying event. In the same way, time records the presumptive sentence attached to the conviction structure that survived the process, not the whole path of charging and negotiation that produced it. The dataset sees the final sentencing record more clearly than it sees the bargaining that shaped the record.

Explorer Task: Translating time Through the Two-Thirds Rule

Here is a task a student could undertake with the Explorer to understand the central lesson of this module: the difference between correct arithmetic and correct interpretation.

The task asks students to filter to a familiar offense category, read the median time, translate that median through the two-thirds rule, and then ask whether that calculation applied to the right population. The task is intentionally simple. The arithmetic is not the point. The point is to teach students that a calculation can be mathematically correct while being substantively wrong because it was applied to cases the rule does not describe.

This task uses terroristic threats, reached through the filter chain $\text{moc1} = A$ and then $\text{moc2} = 9$. This offense category is chosen for three reasons. First, students have already encountered it in earlier modules, so the classification is familiar. Second, its sentence distribution was previously used as a relatively compact example, making it less likely that the lab will be dominated by extreme

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outliers. Third, it gives students a clean setting in which to see how disposition changes the population under analysis.

What students should learn is precise: the two-thirds rule applies only to executed sentences. If students calculate two-thirds of the median time for all terroristic-threats cases, they have performed correct arithmetic on the wrong denominator. Only after filtering to executed prison sentences does the calculation describe the population to which the rule actually applies.

To begin, students should clear the Explorer and apply the filters $moc1 = A$ and $moc2 = 9$. They should then read the descriptive statistics for time, beginning with N. N is the denominator for every claim that follows. Students should then record the median of time.

If the median is 15 months, the two-thirds translation is simple. Two-thirds of 15 months is 10 months. The remaining one-third is 5 months. Under an executed sentence, that would ordinarily mean 10 months in prison and 5 months on supervised release (Minn. Stat. § 244.101, 2024). At this point, the arithmetic is correct.

But the interpretive problem is now visible. The filtered terroristic-threats view includes all sentencing records in that offense category, not only executed prison sentences. Many of those sentences may have been stayed. For stayed sentences, the two-thirds prison calculation describes nothing. It is a prison-sentence calculation applied to a group that includes non-prison sentences.

The next step is therefore to add a disposition filter. Students should keep the terroristic-threats filters in place and add *prison* = Yes, using the column that identifies whether the case resulted in an executed prison sentence. This creates the population to which the two-thirds rule is relevant.

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The JSON version of the lab reports two expected findings. First, N collapses sharply when the filter is limited to executed prison sentences. Second, the executed-only median rises to about 27 months, compared with about 15 months in the all-cases view. The cases that result in prison are selected. They are not a random sample of all terroristic-threats cases. They are likely to be more serious cases, cases involving higher criminal-history scores, cases with aggravating features, or cases that differ in other recorded and unrecorded ways.

That shift is the finding. Filtering to the population for which the two-thirds rule actually applies changes the statistic. The student who translated the all-cases median did not fail at arithmetic. They failed at population definition.

The desired takeaway is this sentence: a correct calculation applied to the wrong population produces a confidently wrong answer.

What the Dataset Can and Cannot Tell You

The dataset can help students improve the calculation. If the prison column records whether a sentence was executed, then students can filter to executed cases before applying the two-thirds rule. That creates a better descriptive claim: among executed prison sentences in a given filtered view, the median sentence can be translated into an ordinary imprisonment term and supervised release term under Minnesota's two-thirds structure.

But the dataset still has a ceiling. It does not fully reconstruct punishment experienced. For stayed sentences, it may not fully report local jail time served as a condition of probation. For executed sentences, it may not show whether supervised release was later revoked. It may not show whether disciplinary violations extended confinement. It does not show every charge initially filed,

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every dismissed count, every plea negotiation, or every unrecorded fact that shaped the conviction offense.

Those omissions are not technical trivia. They mark the boundary between a sentencing dataset and a full account of punishment. The time column is an excellent measure of guideline structure. It is useful for comparing presumptive sentence duration across offense categories, years, criminal-history scores, and recorded groups. It is a much weaker proxy for actual time served.

This distinction will matter even more in later modules. If one group has a higher median time than another, the dataset has identified a difference in presumptive sentence duration. It has not automatically identified a difference in actual prison months served. The same caution applies to offense categories, years, sex, race, and any other comparison built from time.

Summary

The core claims of this module can be restated compactly. The variable *time* records presumptive sentence duration in months. It is not, by itself, a record of prison time served. A presumptive sentence combines disposition and duration: whether the Guidelines recommend prison or a stay, and for how long. Many felony sentences are stayed, meaning no state-prison term is immediately executed. Even when a sentence is executed, Minnesota's two-thirds rule ordinarily divides it into a prison term and a supervised release term. Departures may move the sentence away from the presumptive recommendation, and plea bargaining may shape the conviction offense before the grid ever produces a number.

Thus, if *time* = 48, the honest answer to "how long in prison?" is: it depends. If the sentence was stayed, the answer may be zero months in state prison. If the sentence was executed and followed

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the ordinary statutory path, the prison portion would usually be 32 months. If release was revoked or disciplinary confinement altered the path, actual confinement could move closer to 48 months.

One number can point toward several realities. From this point forward, time should be read as the system's presumptive sentence duration: a powerful measure of guideline structure and a hazardous shortcut for punishment actually experienced.

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Unit III — Reading Patterns Responsibly

Module 8: Change Over Time

`sentyear` as a variable; volume and median-sentence trends 2001–2019; anchoring trends to policy events — centrally the ****2016 Minnesota drug sentencing reform**** (verify effective dates and which offenses; the dataset holds ~3 post-reform years). The lesson in method: a discontinuity that lines up with a law is **suggestive**, not proof — composition changes too.

Data anchor to hunt: drug-offense median `time` and volume before/after 2016.

Platform prerequisite: this is the module where the absence of line charts hurts most — schedule after visualization ships, or lean on small year-by-year tables.

Module 9: Holding Things Constant

The course's methodological hinge. Comparing groups fairly: composition effects, the filter-as-control move ("how much of the pattern survives when I hold this constant?"),

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and — if the data yields one — a real ****Simpson's-paradox**** example where a gap reverses inside subgroups. Lab: take a raw gap, add one control filter at a time, narrate what happens to the gap at each step.

Data anchor to hunt: a genuine sign-reversal or major attenuation example; offense-mix differences between two groups are the most likely source.

Module 10: Small Numbers, Big Mistakes

Expansion of check-N-first: rare offenses, single-county single-year cells, the volatility of small-cell means across adjacent years (same filter, consecutive years, wildly different means — nothing changed but noise). The fireworks three return as old friends.

Data anchor: one rare offense whose year-to-year mean swings absurdly.

Module 11: Sex and Sentencing

The gender gap as the **methodological dress rehearsal** for Unit IV: women are rarer in the system and sentenced more shortly — now apply Module 9's toolkit. Does the gap survive holding offense family constant? History score? What remains, and what **can't** this data say about why? The pattern is among the most robust in the sentencing literature, which makes it the right place to practice disparity reasoning with lower interpretive stakes.

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Builds directly on the Module-3 base-rate example.

Learning objectives. By the end, students can (1) distinguish a raw gap from a conditional gap; (2) execute a stratified comparison in the Explorer using offense family and criminal-history score; (3) articulate what a residual gap does and does not establish; (4) name at least three candidate explanations the dataset cannot adjudicate between.

1. Opening hook: two numbers that can't both be the headline (~3 min)

Present the raw pattern without commentary: women are roughly 15–20% of sentencing records, and their mean pronounced time is substantially shorter than men's. Pose the framing question students should now anticipate: *is this one fact or two?* Fewer women in the data and shorter sentences for women are separate observations, sitting at different funnel depths. Explicit callback to Module 3: recall that men will out-count women on nearly everything simply because of group size — so every comparison in this module must be in percentages and group-wise means, never raw counts.

2. Why gender first, race later (~2 min)

Name the pedagogical move openly. The gender gap is among the most consistently replicated findings in the sentencing literature (Daly & Bordt; Starr's federal work; MSGC's own monitoring reports), which makes it a low-stakes arena to practice the reasoning: students can focus on *method* without the interpretive weight Unit IV will carry. State the promise: every move made in this module will be repeated there, with the same discipline.

3. The naive comparison and why it's not enough (~4 min)

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Run the unconditioned crosstab: sex $\times$ mean time. Establish the gap's size. Then rehearse the objection students should raise themselves by now: men and women may not be *in* the data for the same offenses. If women's records cluster in lower-severity offense families (property, drug) and men's in person offenses, the raw gap partly reflects *composition*, not *treatment*. This is Simpson's-paradox territory — flag the name, keep the math informal.

4. First conditioning: hold offense family constant (~5 min)

Drill into moc1 families one at a time and re-run sex $\times$ mean time within each. Two findings to surface: (a) the composition story is partly right — the mix differs; (b) the gap *shrinks but does not vanish* within most families. Introduce the vocabulary formally: the *within-stratum* gap, the *residual* gap.

5. Second conditioning: hold criminal history constant (~5 min)

Repeat within history bands (0, 1–2, 3+), ideally within a single offense family so students see two-way conditioning. Key observation: women's history scores skew lower, so some of the raw gap flows through history — which is itself an interesting fact, since history scores encode *prior* system contact, meaning earlier funnel stages are baked into this variable. The gap narrows again. Something remains.

6. The hard part: what the residual gap can't tell you (~5 min)

The module's payload. The residual gap is a *description that has survived two adjustments* — it is still not an explanation. Walk the candidate explanations the data cannot separate:

Judicial leniency (differential treatment at sentencing)

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Charging and plea decisions upstream — the funnel again; identical conduct may enter the data as different offense codes by sex

Unrecorded legally relevant factors — role in the offense, weapon use, victim injury within the same moc code

Unrecorded extralegal factors — caregiving responsibilities, which courts may weigh through departures (Minnesota's dispositional departures are a natural pointer here; MSGC reports show women receive mitigated dispositional departures at higher rates)

Selection into the dataset itself — if marginal female cases are diverted pre-felony, the women who remain are non-random

Close with the discipline restated: the Explorer can shrink a gap; it cannot assign a cause.

Adjudicating between these requires data the monitoring system was never designed to collect (Module 1's found-data lesson, returned).

7. Checkpoint + bridge to Unit IV (~2–3 min)

Short quiz (see tasks below), then the bridge: name the exact sequence just performed — raw gap → composition check → conditioning on offense → conditioning on history → residual gap → enumerate unobservables — and state that Unit IV applies this sequence verbatim, where the stakes and the temptation to skip steps are both higher.

Interactive task ideas within the Explorer

Core sequence (in-module)

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Reproduce the raw gap. Clear history, run column summary on sex (counts and percentages), then crosstab sex × mean time. Record both numbers — reinforces that "rarer" and "shorter" are two findings.

Composition audit. Crosstab sex × moc1 in percentages. Question: which offense family shows the largest male–female difference in *share of that group's records*? (Base-rate reasoning applied.)

One-family conditioning. Filter to drug offenses; re-run sex × mean time. Quiz: did the gap grow, shrink, or hold? Repeat for person offenses; compare.

Two-way conditioning. Within one offense family, filter history == 0 and compare sex means; then history >= 3. Question: is the gap larger among first-timers or repeat offenders — and write one sentence on why that's interesting *without* claiming to know the cause.

The trap task. Ask students to find a stratum where the gap *reverses* or nearly vanishes, then explain why one cell doesn't overturn the overall pattern (small-n cells, mode/mean instability — Module 3 callback).

"What can't this view say?" Free-response: given the final conditioned view, list two explanations consistent with it that the dataset cannot distinguish. Graded on enumerating unobservables, not on picking a favorite.

Extension / optional

Median vs. mean check. Re-run a key comparison with medians. If the gap changes size, connect to skew and the mean–median lesson from Module 3.

Time trend. Crosstab sentyear × mean time split by sex: has the gap narrowed since 2001? Caution prompt: guideline changes over time are a confound.

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Severity-weight probe. Compare twweight distributions by sex within an offense family — a proxy for whether men's cases carry more additional counts, i.e., composition hiding *inside* a stratum.

Predict-then-check. Before running task 3, students write a predicted direction and magnitude; the Explorer adjudicates. Builds the habit of registering expectations before looking.

Simpson's construction challenge. Advanced: find (or approximate) a pair of strata where within-group gaps point one way but the pooled gap points differently, and explain the aggregation mechanics.

Write the honest headline. Students draft a one-sentence, publication-ready summary of the module's finding that a methodologist couldn't object to — practicing the descriptive register ("women's sentences average X months shorter within the same offense family and history band") over the causal one ("courts favor women").

Unit IV — The Big Question

Module 12: Race & Sentencing Disparity

Race and Sentencing Disparity

A racial gap in sentencing appears to invite an immediate conclusion. The number looks political, moral, and empirical all at once. If Black defendants receive longer sentences than White defendants in the unfiltered dataset, it is tempting to say that the sentencing system discriminates.

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This module is the point toward which the course has been moving, and for that reason it requires the greatest restraint. Students now have the tools to split sentence length by any group in the data, including race. The Explorer can produce a raw racial crosstab in seconds. The harder task is learning what that table can mean, what it cannot mean, and how to write the result without allowing the claim to outrun the evidence.

The governing distinction is between disparity and discrimination. Descriptive data can document disparity. It cannot, **by itself**, prove discrimination (National Research Council [NRC], 2004). That distinction is not a minor vocabulary preference. It is the boundary between a measured pattern and a causal claim.

The criminal-justice funnel should remain visible throughout the analysis. Each row in the dataset represents a felony sentencing record that has already passed through policing, charging, plea negotiation, conviction, and sentencing. The dataset observes the bottom of the funnel. It does not observe everyone who offended, everyone who was stopped, everyone who was investigated, everyone who was charged, or everyone whose case was dismissed. By the time a case becomes a row, upstream decisions have already shaped who appears in the data. That fact does not make the dataset useless. It makes the dataset bounded.

Disparity and Discrimination

Disparity is a measured difference in outcomes between groups. It is descriptive rather than causal. A statement such as “Black defendants in this dataset receive longer median sentences than White defendants” identifies a disparity in the sentencing records; it does not, by itself, establish why that difference exists or whether it reflects discrimination. Mitchell and MacKenzie (2004) make this distinction directly, explaining that racial and ethnic disparity indicates

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differences between groups on an outcome, whereas discrimination refers to differences that stem from biased treatment.

Discrimination is a causal claim. It means that race itself changed how the case was handled, holding everything else constant. The cleanest version of the question is counterfactual: would this defendant's sentence have been different if only the defendant's race were different, with the same offense, the same conduct, the same criminal history, the same case facts, and the same institutional path? The phrase "all else equal" carries the entire difficulty (Baumer, 2013; NRC, 2004).

Criminal-justice scholarship has long treated this distinction as essential. A disparity may signal that discrimination is possible, but establishing discrimination requires showing that the difference reflects unequal treatment rather than other differences between groups, cases, or institutional pathways (Garland et al., 2008). The pattern is real. The explanation remains complex.

This distinction gives students their first rule for the module: say "disparity" when the claim is descriptive, and reserve "discrimination" for evidence that supports a causal interpretation.

The Raw Crosstab

The first analysis is usually the simplest one. Split the outcome by race, raw and unadjusted. In the Explorer, this means examining the median of time and crosstabbing it by race without filtering the dataset.

Here is a task a student could undertake with the Explorer to understand the raw racial gap. The student should clear the Explorer, summarize the median of time, and then crosstab time by race. No filters should be applied. This task was chosen because it reproduces the first comparison that many analysts would instinctively run: a direct racial comparison in sentence length. What the

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student should notice is that Black defendants' median sentence appears to run a few months longer than White defendants' median sentence in the unconditioned view. What the task is meant to teach is that the raw gap is real as a description, but incomplete as an explanation.

A raw racial gap pools together cases that may differ in nearly every legally relevant way: offense family, offense severity, criminal-history score, disposition, and case pathway. Sentencing research repeatedly emphasizes that these legal and administrative factors structure sentence length and must be interpreted before group differences are explained (Ulmer, 2012). The unadjusted crosstab therefore answers only a narrow question: what is the difference in sentence length between racial groups among all records in the current view? It does not answer whether similar cases were treated differently.

Simpson's Trap

The danger of the raw crosstab is not merely that it may exaggerate a gap. In some cases, an aggregate comparison can reverse when the data are disaggregated. This is the lesson commonly called Simpson's paradox.

The classic example comes from the study of graduate admissions at the University of California, Berkeley. The aggregate data appeared to show bias against women. Once admissions were examined department by department, the pattern vanished and even slightly reversed, because women had applied more often to more competitive departments (Bickel et al., 1975). Blyth (1972) formalized this kind of reversal as Simpson's paradox.

The lesson for sentencing data is direct. A racial gap measured across mixed-together cases can point the reader toward the wrong interpretation if the groups differ in their composition. One group may appear to receive longer sentences overall because its records are concentrated in offense

categories, severity levels, or criminal-history scores that carry longer presumptive sentences.

When the comparison is broken apart into more comparable groups, the pattern may shrink, disappear, remain, or reverse.

Both results can be true at once. The aggregate comparison describes the pooled population. The conditioned comparison describes a narrower population. The analytical question is not which number is “real.” The question is which population each number describes.

Conditioning the Comparison

The next step is to apply the conditioning logic introduced earlier in the course. A filter looks like a technical operation. In this module, it becomes a methodological question: how much of the racial gap remains when legally relevant composition is held constant?

The conditioning chain should be built deliberately. Students should not throw controls into the analysis all at once and treat the final number as magic. They should add case factors one at a time and watch the gap move. The sequence matters because it shows which part of the case composition is doing the most work.

The principal controls in this module are offense family, offense severity, and criminal-history score. These are legally relevant because sentencing is structured around what the conviction offense is, how serious it is, and how prior criminal history is scored. At the national level, Alfred Blumstein (1982) used a similar decomposition logic to ask how much of the Black-White imprisonment gap could be accounted for by differences in arrests for serious crimes. He found that a large share of the gap could be accounted for in this way, although the unexplained remainder was largest for less serious and drug offenses (Blumstein, 1993). The purpose of the

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method is not to make disparity disappear. It is to separate composition from residual difference as carefully as the data allow.

Here is a task a student could undertake with the Explorer to understand conditioning. The student should begin with the raw crosstab of median time by race, then build an attenuation table by adding controls one at a time. First, the student should filter to assault records using `moc1 == A` and crosstab time by race. Next, the student should add `severity == 4`, the most common assault severity level in the example. Finally, the student should add `history == 0`, limiting the view to records with no prior criminal-history score. This task was chosen because it allows students to watch the raw gap change as the cases become more alike. What the student should notice is that the assault gap remains roughly three months after filtering to the offense family, moves very little after adding severity, and then closes to essentially nothing after adding criminal history. What the task is meant to teach is that a raw gap may be driven largely by case composition, and that the control doing the most work matters for interpretation.

In the assault example, the conditioned result does not support the claim that similar cases in that filtered view received different median sentence lengths by race. Once offense family, severity, and criminal history are held constant, the observed gap essentially closes. The honest claim is narrower and stronger than a sweeping conclusion: among assault records at severity level 4 with criminal-history score 0, the racial gap in median time is essentially zero.

That is not a verdict that discrimination is absent. It is a finding about the current filtered view.

The most important part of the example is that the gap closed chiefly when criminal history was added. That finding suggests that the raw gap was largely composition: the racial groups differed in their mix of criminal-history scores, and that difference mattered for sentence length. But criminal

history is not a neutral object floating outside the system. It is a downstream record of earlier arrests, charges, convictions, and sentencing decisions. The control explains the gap statistically. It does not necessarily cleanse the system causally.

The Residual and Its Limits

Suppose a racial gap survives careful conditioning. That remaining difference is often called the residual. A residual gap is important, but it is not self-interpreting. It may be consistent with discrimination, but it may also reflect unobserved case factors, upstream decisions, selection into the dataset, or some mixture of all three.

The first limitation is unobserved confounding. The Explorer can hold constant only the variables the dataset records. It cannot hold constant the strength of the evidence, the precise conduct beneath the offense code, victim cooperation, attorney quality, local courtroom culture, plea bargaining details, or the facts that made one case more serious than another within the same legal category. If any unmeasured factor is associated with race and sentence length, the residual may absorb that factor and appear as a race gap (Baumer, 2013; Klepper et al., 1983).

This is the lesson from offense classification in another form. An offense code is a label, not the conduct itself. “Same offense” in the Explorer means same recorded code. It does not mean same act.

The second limitation is controlling for the wrong things. Some variables that look like controls are themselves products of earlier discretion. Charge severity, mode of conviction, and parts of criminal-history score are not simply facts about the defendant. They are records produced by police, prosecutors, judges, plea negotiations, and prior case processing. Methodological scholarship warns that adjusting for a variable affected by the process under study can bias the

estimate (Rosenbaum, 1984). Conditioning can also create or distort associations when the variable being conditioned on functions as a collider or selection mechanism (Elwert & Winship, 2014).

In criminal justice, this problem cuts deeply. Rehavi and Starr (2014) found that much of the Black-White gap in federal sentence length traced back to prosecutors' initial charging decisions rather than only to the judge's final sentencing decision. If prosecutors charge similar conduct more severely for one group, then controlling for charge severity may remove the very disparity the analyst is trying to detect. More controls are not always better. A control can clarify. It can also launder upstream inequality out of the estimate.

The assault example illustrates the same caution. Criminal history closed the gap, but criminal history is itself an accumulated downstream record. It may reflect prior offending, prior enforcement, prior charging, prior conviction, and prior sentencing. Conditioning on it may remove legitimate legal composition. It may also absorb the residue of earlier discretionary decisions.

The third limitation is selection into the dataset. The dataset contains only people who reached felony sentencing. That population is not equivalent to everyone who offended, everyone who was arrested, everyone who was charged, or everyone who entered the criminal-justice system. Every earlier stage of the funnel is a filter, and those filters may operate differently by race. When a sample is defined by surviving earlier stages, comparisons inside that sample can be distorted by the selection process that produced it (Bushway et al., 2007; Elwert & Winship, 2014; Knox et al., 2020).

This means that a residual gap does not settle the discrimination question. It also means that a vanished gap does not settle it either. A gap may vanish because the relevant controls account for

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legitimate case composition. It may also vanish because the controls absorbed upstream discrimination, because the dataset failed to measure crucial case facts, or because the population reaching sentencing had already been reshaped before the analysis began.

The Explorer can shrink a gap. It cannot assign a cause.

What the Conditioned Gap Can Say

The conditioned comparison is still valuable. It tells the reader how the descriptive gap changes as the population becomes more comparable. That is a meaningful empirical result. The mistake is treating the result as a verdict.

If the gap survives conditioning, the defensible statement is that a residual disparity remains among records that are similar on the observed controls. The result is consistent with unequal treatment, but it is also consistent with unobserved case factors, upstream selection, and omitted variables. If the gap vanishes, the defensible statement is that the observed disparity is accounted for by the measured composition of the cases, while noting which variables absorbed the gap and why those variables may themselves be downstream of earlier discretion.

In the assault example, the most precise statement is this: among assault records filtered to severity level 4 and criminal-history score 0, the racial gap in median presumptive sentence duration essentially closes. In the unfiltered assault view, the raw gap appears to be largely composition, chiefly criminal-history mix. Because criminal history is itself partly downstream of earlier criminal-justice decisions, this finding does not prove that discrimination is absent.

That sentence is longer than “there is no discrimination.” It is also more honest.

Theories of Racial Disparity

The empirical caution in this module sits inside a larger criminological debate. Scholars have long disagreed about where racial disparity is generated in the criminal-justice funnel.

One major distinction is between differential involvement and differential selection. The differential-involvement view holds that much of the disparity enters near the mouth of the funnel through differences in offending and arrest for serious crimes. Blumstein's national decomposition found that a large share of racial disproportionality in imprisonment could be accounted for by racial differences in arrests for serious crimes (Blumstein, 1982). Hindelang (1978) similarly argued that victimization data, which are less dependent on police discretion than arrest records, broadly matched arrest patterns for serious personal crimes.

The differential-selection view locates disparity in the discretionary valves of the system: policing, charging, plea bargaining, conviction, and sentencing. On this view, racial differences may be generated or amplified as cases move through institutional decision points.

A careful reading of the literature does not require choosing only one mechanism. Sampson and Lauritsen (1997) concluded that overt, systematic bias was not the only or primary driver of racial disparity, but that disadvantages can accumulate indirectly across stages. Tonry's account accepts that differential involvement explains some violent-crime disparity, while arguing that the added disproportion, especially for drug offenses, reflects the foreseeable disparate impact of policy choices such as the War on Drugs (Tonry, 1995; Tonry & Melewski, 2008). Sentencing reform has repeatedly sought racially neutral procedures, but neutrality in formal rules does not eliminate all pathways through which disparity can enter case processing (Spohn, 2000).

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Other theories focus on the mechanisms operating at discretionary stages. Focal concerns theory argues that judges and prosecutors make decisions under time pressure and incomplete information. They rely on assessments of blameworthiness, dangerousness, and practical constraints. Race can enter through these perceptual shortcuts rather than through explicit animus. This helps explain why young Black men may face especially severe treatment even under guideline systems (Steffensmeier et al., 1998; Ulmer, 2012).

Cumulative disadvantage theory treats the funnel as a compounding process. Small disparities at individual stages may accumulate as a case moves from police contact to charging, plea negotiation, conviction, and sentencing. A single-stage sentencing analysis may therefore understate the total system-level disparity because it observes only one point in the chain (Kurlychek & Johnson, 2019; Kutateladze et al., 2014).

Racial threat theory emphasizes local political and demographic context. Punishment intensity may vary with racial composition, perceived threat, and local power relations, meaning that pooled statewide or national comparisons may hide important jurisdictional variation (Blalock, 1967; Bridges & Crutchfield, 1988).

These theories matter because they imply different empirical patterns. Differential involvement predicts that much of the gap will be accounted for by offense mix and legally relevant case factors. Differential selection predicts that gaps may appear or grow at discretionary decision points. Focal concerns theory predicts patterned disparities where decision makers rely on proxies for blameworthiness or risk. Cumulative disadvantage predicts compounding across stages. Racial threat theory predicts local variation. A theory is useful because it tells the analyst where to look.

What Would Better Support a Discrimination Claim

If descriptive sentencing data cannot adjudicate discrimination by itself, the next question is what kind of evidence can. The strongest designs get closer to the counterfactual. They try to manufacture the “all else equal” that ordinary administrative data cannot supply.

Audit and correspondence studies do this by sending otherwise-identical profiles that differ only in racial signal. Pager’s audit study found that a criminal record sharply reduced employment callbacks and that the penalty was especially severe for Black applicants. In one of the study’s most striking findings, a White applicant with a criminal record received more favorable treatment than a Black applicant without one (Pager, 2003). Bertrand and Mullainathan (2004) used a similar correspondence design to show that résumés with White-sounding names received more callbacks than otherwise similar résumés with Black-sounding names.

Natural experiments use institutional assignment processes that approximate randomization. If cases are assigned to judges or juries in a way that is effectively random, researchers can compare outcomes while holding case characteristics constant on average. Studies using these strategies have found evidence of racial bias in bail decisions, variation in judicial treatment of race, and racially patterned conviction outcomes depending on jury composition (Abrams et al., 2012; Anwar et al., 2012; Arnold et al., 2018).

These designs are powerful because they identify treatment differences at a specific node of the funnel. That focus is also their limit. A bail study can identify bias in bail decisions. A jury study can identify effects of jury composition. A correspondence study can identify differential treatment in the setting it tests. None of them, alone, reconstructs the entire criminal-justice chain. The funnel remains a system of linked decisions.

Writing the Finding

This module asks students to practice three writing habits.

First, use the right word. A measured racial gap is a disparity. A claim that race caused unequal treatment is discrimination. The words should not be used interchangeably.

Second, name the population and the controls. A sentence such as “among sentenced felony records, holding offense family, severity, and criminal history constant” is not excess detail. It is the claim’s boundary. It tells the reader what population is being described and which parts of composition were held constant.

Third, report the conditioned result without turning it into a verdict. If the gap remains, it is a residual descriptive gap consistent with several explanations. If the gap closes, say which controls absorbed it. In the assault example, the gap closed chiefly when criminal history was added. That is a finding. It is not a clean bill of health.

Here is a task a student could undertake with the Explorer to practice reproducible interpretation. The student should rebuild the fully conditioned assault view by applying `moc1 == A`, then `severity == 4`, then `history == 0`, and then reading time by race. This task was chosen because it requires the student to reproduce the filter history rather than merely remember the conclusion. What the student should notice is that the fully conditioned gap is essentially zero. What the task is meant to teach is that a reproducible finding still requires interpretive discipline: the raw gap was largely composition, chiefly criminal-history mix, but criminal history is itself downstream of earlier discretion that this cross-section of the funnel cannot see.

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The filter history is the methods section. Anyone can replay it. That reproducibility is precisely what makes a careful disparity finding trustworthy, because it shows what the Explorer did and what the analyst refused to overclaim.

Summary

The core lesson of this module is that racial disparity and racial discrimination are not synonyms. A disparity is a measured difference between groups. Discrimination is a causal claim about unequal treatment. The mistake this module prevents is reading a raw racial crosstab as though it automatically reveals discrimination or, in the opposite direction, reading a vanished conditioned gap as though it proves fairness.

From this point forward, every racial comparison must carry its population, its variables, its controls, and its ceiling. The dataset can show whether a gap appears among felony sentencing records. It can show whether that gap narrows, remains, reverses, or vanishes after conditioning. It can show which measured variables absorb the gap. It cannot, by itself, reconstruct the upstream funnel or assign causal responsibility for every difference it records.

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Module 13: What the Data Can't Tell You — and Who Can

The honest ceiling. How researchers go beyond description (conceptual-only tour: regression, matching, the idea of "controlling for" done formally); how to read a sentencing-disparity claim in the news and locate its evidence type; revisiting the funnel. Every analysis in this course is conditional on who reached sentencing. Closes the interpretive arc opened in Module 1.

Unit V — Capstone (Weeks 14–15)

Module 14: Designing a Descriptive Investigation

Students pose their own answerable question, pre-specify a filter plan, run it, and check N at every node. Implemented, but more testing before writing.

Module 15: Communicating Findings

Writing with statistics: choosing mean vs. median and defending the choice; uncertainty

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language; precise descriptive claims (no causal leakage); a short written report whose appendix is the filter-token chain. Rubric is drawn from Modules 1, 9, 10, and 12.

The Educator Portal

Conclusion